

3 Serre's Theorem

(3.1) Theorem (Serre [1971]). *If Γ is a torsion-free group and Γ' is a subgroup of finite index, then $\text{cd } \Gamma' = \text{cd } \Gamma$.*

PROOF. In view of 2.4a, we need only show that if $\text{cd } \Gamma' < \infty$ then $\text{cd } \Gamma < \infty$. It is possible, as we will indicate below, to give a purely algebraic proof of this result; but we will give instead a topological proof, when will yield important information that will be needed in Chapters IX and X. This topological proof requires the following result of Eilenberg and Ganea [1957], which we will prove later as part of Theorem 7.1: If Γ is a group such that $\text{cd } \Gamma < \infty$, then there exists a finite dimensional $K(\Gamma, 1)$ -complex.

Returning now to the proof of 3.1, we are given that $\text{cd } \Gamma' < \infty$, so there is a finite-dimensional $K(\Gamma', 1)$. Its universal cover X' is then a finite dimensional, contractible, free Γ' -complex. To prove $\text{cd } \Gamma < \infty$, we will construct from X' a finite dimensional, contractible, free Γ -complex X . The construction, which is a straightforward analogue of the co-induction construction for modules, goes as follows:

The underlying set of X is defined by $X = \text{Hom}_{\Gamma'}(\Gamma, X')$, where Γ' acts on Γ by left translation and $\text{Hom}_{\Gamma'}(\ , \)$ denotes maps in the category of left Γ' -sets. Since the right action of Γ on itself commutes with the left action of Γ' on Γ , there is an induced (left) action of Γ on X , given by $(\gamma_0 f)(\gamma) = f(\gamma\gamma_0)$ for $f \in X$, $\gamma, \gamma_0 \in \Gamma$.

If we choose a set of coset representatives $\gamma_1, \dots, \gamma_n$ for $\Gamma' \backslash \Gamma$, then we obtain a bijection

$$\varphi: X \xrightarrow{\sim} \prod_{i=1}^n X',$$

given by evaluation at $\gamma_1, \dots, \gamma_n$. Since the product on the right has a natural CW -structure (with the cells being the products of the cells of the factors), we can use φ to give X a topology and a CW -structure. [Note: The product is to be given the CW -topology, i.e., the "weak topology" with respect to the cells; this agrees with the usual product topology if X' is countable or locally compact, cf. Lundell-Weingram [1969], §II.5. In any case, the two topologies agree on all compact subsets.]

This structure is independent of the choice of coset representatives; for if we replace $\gamma_1, \dots, \gamma_n$ by $\gamma'_1\gamma_1, \dots, \gamma'_n\gamma_n$ ($\gamma'_i \in \Gamma'$), then the new φ is obtained from the old one by composition with the CW -isomorphism

$$\prod_{i=1}^n \gamma'_i: \prod_{i=1}^n X' \rightarrow \prod_{i=1}^n X'.$$

The structure is also independent of the ordering of the cosets.

It now follows that the Γ -action on X preserves the CW -structure.

Indeed, for any $\gamma \in \Gamma$ we have a commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{\gamma} & X \\ \varphi' \searrow & & \searrow \varphi \\ & \prod_{i=1}^n X' & \end{array}$$

where φ is defined via coset representatives $(\gamma_i)_{1 \leq i \leq n}$ and φ' is defined via $(\gamma_i \gamma)_{1 \leq i \leq n}$. Thus X is a well-defined Γ -complex, which is clearly contractible and finite dimensional.

To complete the proof that $\text{cd } \Gamma < \infty$, we will show that Γ acts freely on X . There is a canonical map $X \rightarrow X'$, given by evaluation at $1 \in \Gamma$. This map is Γ' -equivariant and takes cells to cells. Since Γ' acts freely on X' , it follows that Γ acts freely on X . For any cell σ of X , then, we have $\Gamma_\sigma \cap \Gamma' = \{1\}$, hence Γ_σ is finite. But Γ is torsion-free, so these finite isotropy groups Γ_σ are trivial. $\square$

Remark. The interested reader can translate the proof above into a purely algebraic proof. One works directly with projective resolutions, and one does the “co-induction” construction using tensor products of chain complexes rather than cartesian products of CW -complexes. More details can be found in Swan [1969], Theorem 9.2.

EXERCISE

Note that the proof of 3.1 is valid even if Γ has torsion, except for the last sentence of the proof. Consequently, if Γ is a group which contains a subgroup Γ' of finite index such that $\text{cd } \Gamma' < \infty$, then we can construct a contractible, finite dimensional Γ -complex X , with finite isotropy groups Γ_σ . Show that this complex has the following additional property: For every finite subgroup $H \subseteq \Gamma$, the fixed-point set X^H is contractible. [Hint: Show that X , as an H -complex, is isomorphic to the product of $(\Gamma : \Gamma')$ copies of the complex X' , with H acting by permuting the factors according to the (free) right action of H on $\Gamma' \backslash \Gamma$. One can see this either by using the double coset formula for co-induction or simply by direct inspection, using coset representatives of the form $\{\gamma_i h\}_{1 \leq i \leq d, h \in H}$, where $d = (\Gamma : \Gamma')/|H|$. Hence X^H is homeomorphic to the product of d copies of X' .]

4 Resolutions of Finite Type

Our next goal is to study a different sort of finiteness condition, where we require that there be a projective resolution P with each P_i finitely generated. In this section we collect some general facts about such resolutions over an

arbitrary ring R . Then in the next two sections we specialize to resolutions of $\mathbb{Z}$ over $\mathbb{Z}\Gamma$.

We begin by reviewing the theory of finitely presented modules:

(4.1) Proposition. *The following conditions on an R -module M are equivalent:*

- (i) *There is an exact sequence $R^m \rightarrow R^n \rightarrow M \rightarrow 0$ for some integers m, n .*
- (ii) *There is an exact sequence $P_1 \rightarrow P_0 \rightarrow M \rightarrow 0$ for some finitely generated projectives P_0, P_1 .*
- (iii) *M is finitely generated, and for every surjection $\varepsilon: P \rightarrow M$ with P finitely generated and projective, $\ker \varepsilon$ is finitely generated.*

The proof is based on “Schanuel’s lemma”:

(4.2) Lemma. *Let $0 \rightarrow K \rightarrow P \rightarrow M \rightarrow 0$ and $0 \rightarrow K' \rightarrow P' \rightarrow M \rightarrow 0$ be exact sequences with P and P' projective. Then $P \oplus K' \approx P' \oplus K$.*

PROOF. Let Q be the pullback of the given maps

$$\begin{array}{ccc} & P' & \\ & \downarrow \pi' & \\ P & \xrightarrow{\pi} & M, \end{array}$$

i.e., Q is the submodule of $P \times P'$ consisting of those pairs (x, x') such that $\pi(x) = \pi'(x')$. One then verifies easily that there is a commutative diagram

$$\begin{array}{ccccccc} & & 0 & & 0 & & \\ & & \downarrow & & \downarrow & & \\ & & K' & \xlongequal{\quad} & K' & & \\ & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & K & \longrightarrow & Q & \longrightarrow & P' \longrightarrow 0 \\ & & \parallel & & \downarrow & & \downarrow \\ 0 & \longrightarrow & K & \longrightarrow & P & \longrightarrow & M \longrightarrow 0 \\ & & & & \downarrow & & \downarrow \\ & & & & 0 & & 0 \end{array}$$

with exact rows and columns. Since P and P' are projective, the two exact sequences involving Q must split, yielding $P \oplus K' \approx Q \approx P' \oplus K$. $\square$

PROOF OF 4.1. Clearly (iii) $\Rightarrow$ (i) $\Rightarrow$ (ii). To prove (ii) $\Rightarrow$ (iii), note first that M is certainly finitely generated if (ii) holds, since P_0 is finitely generated. Now

apply Schanuel to get $P \oplus \ker\{P_0 \rightarrow M\} \approx P_0 \oplus \ker \varepsilon$. The lefthand side being finitely generated by hypothesis, it follows that $P_0 \oplus \ker \varepsilon$ is finitely generated, hence so is $\ker \varepsilon$. $\square$

M is said to be *finitely presented* if the conditions of 4.1 hold. An exact sequence as in (i) is said to give a *finite presentation* of M with n generators and m relations. Note that the implication (i) $\Rightarrow$ (iii), applied when P is free, reduces to the following well-known fact: If M admits *some* finite presentation, then *every* finite set $x_1, \dots, x_k$ of generators of M has the property that the relations among them (i.e., the k -tuples $(r_1, \dots, r_k)$ such that $\sum r_i x_i = 0$) form a finitely generated submodule of R^k .

It is natural to generalize finite presentation as follows: A resolution or partial resolution (P_i) is said to be of *finite type* if each P_i is finitely generated. A module M is said to be of *type* FP_n ($n \geq 0$) if there is a partial projective resolution $P_n \rightarrow \dots \rightarrow P_0 \rightarrow M \rightarrow 0$ of finite type. Thus the FP_0 condition is simply finite generation, FP_1 is finite presentation, and the conditions $FP_2, FP_3, \dots$ are successive strengthenings of finite presentation.

Generalizing 4.1, we have:

(4.3) Proposition. *For any module M and integer $n \geq 0$ the following conditions are equivalent:*

- (i) *There is a partial resolution $F_n \rightarrow \dots \rightarrow F_0 \rightarrow M \rightarrow 0$ with each F_i free of finite rank.*
- (ii) *M is of type FP_n .*
- (iii) *M is finitely generated, and for every partial projective resolution $P_k \rightarrow \dots \rightarrow P_0 \rightarrow M \rightarrow 0$ of finite type with $k < n$, $\ker\{P_k \rightarrow P_{k-1}\}$ is finitely generated.*

For the proof we will need the following generalization of Schanuel's lemma:

(4.4) Lemma. *Let*

$$0 \rightarrow P_n \rightarrow P_{n-1} \rightarrow \dots \rightarrow P_0 \rightarrow M \rightarrow 0$$

and

$$0 \rightarrow P'_n \rightarrow P'_{n-1} \rightarrow \dots \rightarrow P'_0 \rightarrow M \rightarrow 0$$

be exact sequences with P_i and P'_i projective for $i \leq n-1$. Then

$$P_0 \oplus P'_1 \oplus P_2 \oplus P'_3 \oplus \dots \approx P'_0 \oplus P_1 \oplus P'_2 \oplus P_3 \oplus \dots$$

Consequently, if P_i and P'_i are finitely generated for $i \leq n-1$, then P_n is finitely generated if and only if P'_n is finitely generated.

3. It follows from the isomorphism of 4.4 that P_n is projective if and only if P'_n is projective. Thus 4.4 yields an alternative proof of the implication (i) $\Rightarrow$ (iv) of 2.1.

We will be primarily interested in the case where the conditions of 4.3 hold for all integers $n \geq 0$. This situation is characterized as follows:

(4.5) Proposition. *The following conditions on a module M are equivalent:*

- (i) M admits a free resolution of finite type.
- (ii) M admits a projective resolution of finite type.
- (iii) M is of type FP_n for all integers $n \geq 0$.

PROOF. (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) trivially. If (iii) holds then we can use 4.3(iii) to construct a free resolution of finite type step by step, so (iii) $\Rightarrow$ (i). $\square$

We say that M is of type FP_∞ if these conditions hold.

We close this section by mentioning some useful formal properties which the homology and cohomology functors $\text{Tor}_*^R(M, -)$ and $\text{Ext}_R^*(M, -)$ satisfy when M is of type FP_n . (The reader should keep in mind, during this discussion, the main case of interest: $R = \mathbb{Z}\Gamma$, $M = \mathbb{Z}$. In this case the Tor and Ext functors are simply $H_*(\Gamma, -)$ and $H^*(\Gamma, -)$.)

Note first that the homology functors $\text{Tor}_*^R(M, -)$ always commute with direct limits, in the following sense: Let $\{N_i\}_{i \in I}$ be a direct system of R -modules, where I is a directed set, and let $N = \varinjlim_{i \in I} N_i$. Thus N is the universal target of a compatible family of maps $N_i \rightarrow N$ ($i \in I$). These maps induce a compatible family of abelian group homomorphisms $\text{Tor}_*^R(M, N_i) \rightarrow \text{Tor}_*^R(M, N)$, from which we obtain a map

$$\varphi: \varinjlim_{i \in I} \text{Tor}_*^R(M, N_i) \rightarrow \text{Tor}_*^R(M, N);$$

the assertion, then, is that φ is an isomorphism. This follows directly from the definition of $\text{Tor}_*^R(M, -)$ as $H_*(P \otimes_R -)$, where P is a projective resolution of M , together with the following two facts:

- (a) $P \otimes_R -$ commutes with direct limits (cf. Spanier [1966], 5.1.9);
- (b) $H_*(-)$ commutes with direct limits (cf. Spanier [1966], 4.1.7).

Functors of the form $\mathcal{H}om_R(P, -)$, however, do not in general commute with direct limits, so we cannot expect $\text{Ext}_R^*(M, -)$ to commute with direct limits. But we can prove that this does hold under suitable FP_n hypotheses. For simplicity, we will confine ourselves to the case $n = \infty$.

(4.6) Proposition. *If M is of type FP_∞ then $\text{Ext}_R^*(M, -)$ commutes with direct limits.*

This follows immediately from:

(4.7) Lemma. *If P is a finitely generated projective R -module then $\text{Hom}_R(P, -)$ commutes with direct limits.*

PROOF. The duality isomorphism I.8.3b shows that the functor $\text{Hom}_R(P, -)$ is equivalent to the functor $P^* \otimes_R -$, and we know that the latter commutes with direct limits. $\square$

(See Exercise 4 below for a generalization of 4.7 and an indication of an alternative proof.)

Similarly, one can show that $\text{Tor}_*^R(M, -)$ commutes with direct products if M is of type FP_∞ (whereas $\text{Ext}_R^*(M, -)$ commutes with products for *any* M).

Surprisingly, these formal properties of $\text{Ext}_R^*(M, -)$ and $\text{Tor}_*^R(M, -)$ characterize the FP_∞ property. In fact, one can prove:

(4.8) Theorem. *The following conditions are equivalent:*

- (i) M is of type FP_∞ .
- (ii) $\text{Ext}_R^*(M, -)$ commutes with direct limits.
- (iii) $\text{Tor}_*^R(M, -)$ commutes with direct products.

We omit the proof, since we will not be making serious use of this result. The equivalence of (i) and (iii) was first proved by Bieri and Eckmann [1974]. See Brown [1975a] for the equivalence of (i) and (ii), as well as for a generalization of 4.8. See also Strebel [1976] for further results of this type.

EXERCISES

- Let Γ be a group. Show that $\mathbb{Z}$ is finitely presented as a $\mathbb{Z}\Gamma$ -module if and only if Γ is a finitely generated group. [Hint: Use Exercise 1d of §I.2.]
- Let M' and M'' be modules and let $M = M' \oplus M''$. Show that M is of type FP_n if and only if M' and M'' are of type FP_n . [Hint: Suppose P' and P'' are finite type partial resolutions of M' and M'' of length k , and let $P = P' \oplus P''$. Note that P has finitely generated kernel if and only if P' and P'' do.]
- Let C and C' be non-negative chain complexes of projective modules. If C and C' are homotopy equivalent, show that

$$C_0 \oplus C'_1 \oplus C_2 \oplus \cdots \approx C'_0 \oplus C_1 \oplus C'_2 \oplus \cdots.$$

[Hint: Imitate the alternative proof of 4.4 outlined in Remark 2 following 4.4.]

- If M is a finitely presented R -module, show that $\text{Hom}_R(M, -)$ commutes with direct limits. [Hint: This can be deduced from 4.7. Alternatively, give a direct proof based on the fact that a map from M to any other module can be specified by giving a finite set of elements of the target module which satisfy a certain finite set of relations.]

5 Groups of Type FP_n

We now specialize the theory of §4 to the case $R = \mathbb{Z}\Gamma$, $M = \mathbb{Z}$. We will say that Γ is of type FP_n ($0 \leq n \leq \infty$) if $\mathbb{Z}$ is of type FP_n as a $\mathbb{Z}\Gamma$ -module. Thus every group is of type FP_0 , and it is easy to see that Γ is of type FP_1 if and only if it is finitely generated (cf. Exercise 1 of §4). The FP_2 condition is less well-understood, however. One knows that finitely presented groups Γ are of type FP_2 , for if Y is a finite 2-complex with $\pi_1 Y = \Gamma$ then the cellular chain complex of the universal cover of Y is a partial free resolution of $\mathbb{Z}$ over $\mathbb{Z}\Gamma$ of length 2 and of finite type. [See also exercise 4c of §IV.2 for an algebraic proof.] But it is not known whether the converse is true. See Exercise 3 below for a reformulation of the problem.³

We will not pause now to discuss examples, since there will be plenty of examples of groups of type FP_∞ in §§6, 9, and 11. The reader who wants to see examples for $n < \infty$ of groups of type FP_n but not of type FP_{n+1} should consult Bieri [1976], §2.6, and Stuhler [1980]; see also Exercise 2 below for the case $n = 1$ and Stallings [1963] for the case $n = 2$.

The FP_n conditions behave nicely with respect to subgroups of finite index:

(5.1) Proposition. *Let $\Gamma' \subseteq \Gamma$ be a subgroup of finite index. Then Γ is of type FP_n ($0 \leq n \leq \infty$) if and only if Γ' is of type FP_n .*

PROOF. Any (partial) projective resolution of $\mathbb{Z}$ over $\mathbb{Z}\Gamma$ of finite type can also be regarded as a (partial) projective resolution of $\mathbb{Z}$ over $\mathbb{Z}\Gamma'$, and as such it is still of finite type since $(\Gamma : \Gamma') < \infty$. This proves the “only if” part. Conversely, suppose Γ' is of type FP_n . We will show that the $\mathbb{Z}\Gamma$ -module $\mathbb{Z}$ satisfies condition 4.3(iii). Let P be a partial projective resolution of $\mathbb{Z}$ over $\mathbb{Z}\Gamma$ of finite type and of length $k < n$. Regarding P as a partial resolution of $\mathbb{Z}$ over $\mathbb{Z}\Gamma'$, we can apply 4.3(iii) to conclude that $\ker\{P_k \rightarrow P_{k-1}\}$ is finitely generated over $\mathbb{Z}\Gamma'$. But then $\ker\{P_k \rightarrow P_{k-1}\}$ is certainly finitely generated over $\mathbb{Z}\Gamma$, so 4.3(iii) holds for $\mathbb{Z}$ over $\mathbb{Z}\Gamma$. $\square$

Taking $n = 1$, for example, we recover the well-known (but not completely obvious) fact that Γ is finitely generated if and only if Γ' is finitely generated.

Finally, we record for future reference an important consequence of 4.6:

(5.2) Proposition. *Let Γ be a group of type FP_∞ and let n be an integer such that $H^n(\Gamma, \mathbb{Z}\Gamma) = 0$. Then $H^n(\Gamma, F) = 0$ for all free $\mathbb{Z}\Gamma$ -modules F .*

PROOF. If F is of finite rank then the result follows from the additivity of $H^n(\Gamma, -)$. In the general case, choose a basis $(e_i)_{i \in I}$ for F and note that $F = \varinjlim F_J$, where J ranges over the finite subsets of I and F_J is the

³ For solvable Γ , substantial progress on this question has been made by Bieri and Strebel. For references, see the appendix to the forthcoming second edition of Bieri [1976].

submodule of F generated by the e_i for $i \in J$. Then F_J is free of finite rank; using 4.6 we conclude that $H^n(\Gamma, F) = \varinjlim H^n(\Gamma, F_J) = 0$. $\square$

EXERCISES

1. Let Γ be a group of type FP_n and let M be a Γ -module which is finitely generated as an abelian group. Show that $H_i(\Gamma, M)$ and $H^i(\Gamma, M)$ are finitely generated abelian groups for $i \leq n$.
2. Let Γ be an amalgamation $\Gamma_1 *_A \Gamma_2$ where Γ_1 and Γ_2 are free of finite rank and A is free of infinite rank. Show that Γ is finitely generated but not of type FP_2 . In particular, Γ is not finitely presented. [Hint: Use the Mayer-Vietoris sequence to show that $H_2 \Gamma$ is not finitely generated.]
3. (a) If Γ is of type FP_2 and N is a perfect normal subgroup (i.e., a normal subgroup such that $N = [N, N]$), prove that Γ/N is of type FP_2 . [Hint: Apply the functor $(\)_N$ to a partial resolution of $\mathbb{Z}$ over $\mathbb{Z}\Gamma$ of length 2 and finite type; the resulting complex will still be acyclic in dimension one because its one-dimensional homology is $H_1 N = 0$.]
 (b) Let $\Gamma = F/R$ where F is a finitely generated free group. Prove that the following conditions are equivalent:
 - (i) Γ is of type FP_2 .
 - (ii) The relation module R_{ab} is a finitely generated Γ -module.
 - (iii) $\Gamma \approx \tilde{\Gamma}/N$, where $\tilde{\Gamma}$ is finitely presented and N is a perfect normal subgroup.

[Hint: For (i) $\Rightarrow$ (ii) use II.5.4.]

Remark. In view of the equivalence of (i) and (iii), we can reformulate as follows the question as to whether every group of type FP_2 is finitely presented: Is a perfect normal subgroup of a finitely presented group necessarily finitely generated as a normal subgroup?

4. For any group Γ , note that the cohomology groups $H^*(\Gamma, \mathbb{Z}\Gamma)$ have a canonical structure of *right* Γ -module. Indeed, the coefficient module $\mathbb{Z}\Gamma$, which is thought of as a left module for the purpose of defining $H^*(\Gamma, \mathbb{Z}\Gamma)$, also admits a right Γ -action (by right translation) which commutes with the left action; this right action induces a right action of Γ on the cohomology groups $H^*(\Gamma, \mathbb{Z}\Gamma)$. In this exercise you will prove two “universal coefficient” formulas involving these right Γ -modules.

- (a) If Γ is a group of type FP_∞ , prove that

$$H^*(\Gamma, F) \approx H^*(\Gamma, \mathbb{Z}\Gamma) \otimes_{\mathbb{Z}\Gamma} F$$

for any flat $\mathbb{Z}\Gamma$ -module F . [Hint: Use I.8.3b to rewrite $\mathcal{H}om_\Gamma(P, F)$ as a tensor product, where P is a projective resolution of finite type.]

- (b) If Γ is of type FP_∞ and Q is an injective $\mathbb{Z}\Gamma$ -module, prove that

$$H_*(\Gamma, Q) \approx \mathcal{H}om_\Gamma(H^*(\Gamma, \mathbb{Z}\Gamma), Q).$$

[Here we should take Q to be a *right* Γ -module so that the $\mathcal{H}om$ makes sense.]

5. Let Γ be a free product $\Gamma_1 * \Gamma_2$, where Γ_1 and Γ_2 are infinite and of type FP_∞ . For any integer i show that there is a map of (right) Γ -modules

$$H^i(\Gamma, \mathbb{Z}\Gamma) \rightarrow \text{Ind}_{\Gamma_1}^\Gamma H^i(\Gamma_1, \mathbb{Z}\Gamma_1) \oplus \text{Ind}_{\Gamma_2}^\Gamma H^i(\Gamma_2, \mathbb{Z}\Gamma_2)$$

which is an isomorphism if $i > 1$ and an epimorphism if $i = 1$ with kernel a free $\mathbb{Z}\Gamma$ -module of rank 1. [Hint: Use the Mayer–Vietoris sequence, and apply exercise 4a to compute $H^*(\Gamma_j, \mathbb{Z}\Gamma_j)$, $j = 1, 2$.]

6 Groups of Type *FP* and *FL*

We now combine the two types of finiteness conditions which we have considered in this chapter. A resolution is said to be *finite* if it is both of finite type and of finite length. A group Γ is said to be of type *FP* if $\mathbb{Z}$ admits a finite projective resolution over $\mathbb{Z}\Gamma$.

(6.1) Proposition. Γ is of type *FP* if and only if (i) $\text{cd } \Gamma < \infty$ and (ii) Γ is of type FP_∞ .

PROOF. The “only if” part is obvious. Conversely, if $\text{cd } \Gamma < \infty$ and Γ is of type FP_∞ , then we can construct a finite resolution as follows: Take a partial resolution $P_{n-1} \rightarrow \cdots \rightarrow P_0 \rightarrow \mathbb{Z} \rightarrow 0$ of finite type, where $n = \text{cd } \Gamma$, and let $P_n = \ker\{P_{n-1} \rightarrow P_{n-2}\}$. Then P_n is projective (2.1(iv)) and finitely generated (4.3(iii)), so we have a finite projective resolution

$$0 \rightarrow P_n \rightarrow \cdots \rightarrow P_0 \rightarrow \mathbb{Z} \rightarrow 0. \quad \square$$

Note that it would have been enough in the proof above to assume that Γ was of type FP_n instead of FP_∞ , where $n = \text{cd } \Gamma$. It follows, for instance, that a finitely presented group Γ with $\text{cd } \Gamma = 2$ is of type *FP*. Note also that the partial resolution $(P_i)_{i \leq n-1}$ above could have been taken free. Hence if Γ is of type *FP* then there is a finite projective resolution

$$(6.2) \quad 0 \rightarrow P \rightarrow F_{n-1} \rightarrow \cdots \rightarrow F_0 \rightarrow \mathbb{Z} \rightarrow 0$$

with each F_i free. (See also exercise 2 below.) But there is no reason to expect to be able to take P free. Thus, for the first time in this book there really seems to be a difference between what can be done with projective resolutions and what can be done using only free resolutions.

We are therefore led to introduce a still stronger finiteness condition: Γ is of type *FL* if $\mathbb{Z}$ admits a finite free resolution over $\mathbb{Z}\Gamma$. [Warning: It is a common mistake to assume that “*FL*” stands for “finite length,” and thereby to confuse the *FL* property with the much weaker property of having finite cohomological dimension. In fact, the “*L*” in “*FL*” stands for “libre,” not for “length.” One also finds “*FF*” (“finite free resolution”) in the literature instead of “*FL*.”]

It is obvious how to use topology to obtain examples of groups of type *FL*:

(6.3) Proposition. *If there exists a $K(\Gamma, 1)$ which is a finite complex, then Γ is of type *FL*.*

(We will see in §7 that the converse of this is also true, at least if we assume that Γ is finitely presented.)

Looking at the examples in §2, we immediately deduce the following examples of groups of type *FL*: free groups of finite rank; surface groups; finitely generated one-relator groups whose relator is not a power; and free abelian groups of finite rank. With a little more work (cf. exercise 8 below), one can also show that torsion-free, finitely generated, nilpotent groups are of type *FL*. We will reprove this result and give many additional examples of groups of type *FL* in §9.

The *FP* property also admits a topological interpretation, for which we need the following notion: A space Y is *finitely dominated* if there is a finite complex K such that Y is a retract of K in the homotopy category (i.e., we require maps $i: Y \rightarrow K$ and $r: K \rightarrow Y$ with $ri \simeq \text{id}_Y$).

(6.4) Proposition. *If there exists a finitely dominated $K(\Gamma, 1)$, then Γ is of type *FP*.*

(Again the converse is also true, and will be proved in §7, provided Γ is finitely presented.)

We will not be making any use of this result, so we confine ourselves to a brief sketch of the proof:

Let Y be a $K(\Gamma, 1)$ -complex dominated by a finite complex K . One can choose K so that the maps $Y \rightleftarrows K$ induce π_1 -isomorphisms. Letting $\tilde{Y}$ and $\tilde{K}$ be the universal covers, one deduces that the cellular chain complex $C(\tilde{Y})$ is a retract of $C(\tilde{K})$ in the homotopy category of chain complexes over $\mathbb{Z}\Gamma$. Since $C(\tilde{Y})$ is a free resolution of $\mathbb{Z}$ and $C(\tilde{K})$ is a finite free complex, it follows that $H^*(\Gamma, -)$ commutes with direct limits and that $H^i(\Gamma, -) = 0$ for $i > \dim K$. In view of 4.8, this implies that Γ is of type *FP*. $\square$

Remark. Propositions 6.3 and 6.4 and their converses are special cases of the following result due to Wall [1965, 1966]: Let Y be a connected *CW*-complex whose fundamental group π is finitely presented, and let C be the chain complex of the universal cover of Y , regarded as a complex of π -modules. Then (a) Y is homotopy equivalent to a finite *CW*-complex if and only if C is homotopy equivalent to a finite free chain complex; and (b) Y is finitely dominated if and only if C is homotopy equivalent to a finite projective chain complex.

Having carefully explained the difference between the *FP* condition and the *FL* condition, we are now forced to admit that there are no known

examples of groups of type FP which are not of type FL . [From the topological point of view, we have the following situation: Although there are plenty of known examples of finitely dominated spaces which do not have the homotopy type of a finite complex, none of these known examples are $K(\Gamma, 1)$'s.]

In order to better appreciate the problem, let's see what the obstruction is to proving that a group of type FP is of type FL . Suppose Γ is of type FP , and choose a finite projective resolution as in 6.2 which is free except in the top dimension. The projective P which occurs in the top dimension might have the property that $P \oplus F$ is free for some free module F of finite rank. In this case P is said to be *stably free*, and we can modify 6.2 exactly as in the proof of 2.6 to obtain a finite free resolution. Conversely, if there exists a finite free resolution, then we can compare it to 6.2 via 4.4 to deduce that P is stably free. [Note: The two resolutions might not have the same length, but 4.4 is still applicable; for we can extend the shorter resolution by zeroes.] Thus we have:

(6.5) Proposition. *Let Γ be a group of type FP and let $0 \rightarrow P \rightarrow F_{n-1} \rightarrow \cdots \rightarrow F_0 \rightarrow \mathbb{Z} \rightarrow 0$ be a finite projective resolution of $\mathbb{Z}$ over $\mathbb{Z}\Gamma$ with each F_i free. Then Γ is of type FL if and only if P is stably free.*

Thus the question as to whether there exist groups of type FP which are not of type FL has led to a more fundamental question: Do there exist finitely generated projectives which are not stably free? Over a general ring the answer is certainly "yes," and there are even known examples over integral group rings $\mathbb{Z}\Gamma$, the simplest example being with $\Gamma = \mathbb{Z}_{23}$ (cf. Milnor [1971], §3). The surprising fact, however, is that there are no known examples with Γ *torsion-free*, and a group of type FP is necessarily torsion-free by 2.5.

We remark, finally, that there do exist resolutions as in 6.2 in which P is not free. The first such example was given by Dunwoody [1972], with Γ equal to the trefoil group. In all of the known examples, however, the group Γ is known to be of type FL (and hence P is stably free).

In spite of this lack of examples, we will see later (cf. IX.6.4 and Remark 1 following its proof) that there are concrete results whose proofs require that we consider the FP property, and not just the FL property. One reason for this is that we can prove the following result about groups of type FP , the analogue of which for groups of type FL is not known:

(6.6) Proposition. *Let Γ be a torsion-free group and Γ' a subgroup of finite index. Then Γ is of type FP if and only if Γ' is of type FP .*

PROOF. This follows from 3.1 and 5.1. □

In particular, if Γ is a torsion-free group which contains a subgroup of finite index which is of type FL , then we know from 6.6 that Γ is of type FP , even though we don't know that Γ is of type FL .

We close this section by discussing some special features of the top-dimensional cohomology of a group of type FP . First we note the following improvement of 2.3:

(6.7) Proposition. *If Γ is of type FP then $\text{cd } \Gamma = \max\{n: H^n(\Gamma, \mathbb{Z}\Gamma) \neq 0\}$.*

PROOF. This follows from 2.3 and 5.2. $\square$

Recall that the cohomology groups $H^i(\Gamma, \mathbb{Z}\Gamma)$ admit a canonical *right Γ -module structure* (cf. exercise 4 of §5). For any (left) Γ -module M , we can therefore form the tensor product $H^i(\Gamma, \mathbb{Z}\Gamma) \otimes_{\mathbb{Z}\Gamma} M$, and there is a canonical map

$$\varphi: H^*(\Gamma, \mathbb{Z}\Gamma) \otimes_{\mathbb{Z}\Gamma} M \rightarrow H^*(\Gamma, M),$$

defined as follows on the cochain level: Let P be a projective resolution of $\mathbb{Z}$ over $\mathbb{Z}\Gamma$; given a cochain $u \in \mathcal{H}om_{\Gamma}(P, \mathbb{Z}\Gamma)$ and an element $m \in M$, we send $u \otimes m$ to the cochain $x \mapsto u(x)m$ ($x \in P$) in $\mathcal{H}om_{\Gamma}(P, M)$.

We can now prove the following “universal coefficient theorem” for the top-dimensional cohomology of a group of type FP :

(6.8) Proposition. *If Γ is of type FP and $n = \text{cd } \Gamma$, then*

$$\varphi: H^n(\Gamma, \mathbb{Z}\Gamma) \otimes_{\mathbb{Z}\Gamma} M \rightarrow H^n(\Gamma, M)$$

is an isomorphism for all Γ -modules M .

PROOF. We will give two proofs, both of which are instructive.

Proof 1: Regard φ as a natural transformation between functors of M . Since both functors are right exact, it suffices to prove that φ is an isomorphism when M is free. [The general case is then obtained by considering an exact sequence $F' \rightarrow F \rightarrow M \rightarrow 0$ with F and F' free.] Since both functors are additive and commute with direct limits, it suffices to consider the case where M is free of rank 1, i.e., we may assume $M = \mathbb{Z}\Gamma$. But in this case $\varphi: H^n(\Gamma, \mathbb{Z}\Gamma) \otimes_{\mathbb{Z}\Gamma} \mathbb{Z}\Gamma \rightarrow H^n(\Gamma, \mathbb{Z}\Gamma)$ is simply the well-known isomorphism $u \otimes r \mapsto ur$ (with inverse $u \mapsto u \otimes 1$).

Proof 2: Let P be a finite projective resolution of $\mathbb{Z}$ over $\mathbb{Z}\Gamma$ of length n and let $\bar{P}$ be the dual complex $\mathcal{H}om_{\Gamma}(P, \mathbb{Z}\Gamma)$ of right Γ -modules. Then $H^*(\Gamma, \mathbb{Z}\Gamma) \approx H^*(\bar{P})$. In particular, we have an exact sequence

$$\bar{P}^{n-1} \rightarrow \bar{P}^n \rightarrow H^n(\Gamma, \mathbb{Z}\Gamma) \rightarrow 0$$

for the top-dimensional cohomology module. Tensoring with M and applying the duality isomorphism I.8.3b, we obtain the diagram

$$\begin{array}{ccccccc}
 \bar{P}^{n-1} \otimes_{\mathbb{Z}\Gamma} M & \longrightarrow & \bar{P}^n \otimes_{\mathbb{Z}\Gamma} M & \longrightarrow & H^n(\Gamma, \mathbb{Z}\Gamma) \otimes_{\mathbb{Z}\Gamma} M & \longrightarrow & 0 \\
 \downarrow \approx & & \downarrow \approx & & \downarrow \varphi & & \\
 \text{Hom}_{\Gamma}(P_{n-1}, M) & \longrightarrow & \text{Hom}_{\Gamma}(P_n, M) & \longrightarrow & H^n(\Gamma, M) & \longrightarrow & 0.
 \end{array}$$

which commutes and has exact rows. It follows that φ is an isomorphism. $\square$

The isomorphism of 6.8 can be written in the following suggestive way: Let D be the Γ -module $H^n(\Gamma, \mathbb{Z}\Gamma)$, so that $H^n(\Gamma, \mathbb{Z}\Gamma) \otimes_{\mathbb{Z}\Gamma} M = D \otimes_{\mathbb{Z}\Gamma} M = (D \otimes M)_{\Gamma} = H_0(\Gamma, D \otimes M)$, where $D \otimes M = D \otimes_{\mathbb{Z}} M$ with the diagonal Γ -action. [Note: Since D is a right module and M is a left module, the diagonal action is defined by $\gamma \cdot (d \otimes m) = d\gamma^{-1} \otimes \gamma m$ for $\gamma \in \Gamma$, $d \in D$, $m \in M$.] Consequently, 6.8 can be viewed as an isomorphism

$$(6.9) \quad H^n(\Gamma, M) \approx H_0(\Gamma, D \otimes M).$$

We will return to this point of view in §10.

EXERCISES

1. If Γ is of type *FL* and $\text{cd } \Gamma = n$, show that $\mathbb{Z}$ admits a finite free resolution over $\mathbb{Z}\Gamma$ of length n .
2. Let Γ be of type *FP* and let m be an arbitrary integer ≥ 0 . Prove that $\mathbb{Z}$ admits a finite projective resolution P over $\mathbb{Z}\Gamma$ such that P_i is free for $i \neq m$. Moreover, P can be taken to have length equal to $\max\{m, \text{cd } \Gamma\}$. [Hint: Given a resolution, you can modify it by taking the direct sum with a complex of the form $0 \rightarrow Q \xrightarrow{\text{id}} Q \rightarrow 0$.]
3. If Γ is of type *FP* and $n = \text{cd } \Gamma$, show that $H^n(\Gamma, \mathbb{Z}\Gamma)$ is a finitely generated Γ -module.
- *4. Let Γ be of type *FP*, let $n = \text{cd } \Gamma$, and suppose $H^n(\Gamma, \mathbb{Z}\Gamma)$ is finitely generated as an abelian group. If $\Gamma' \subset \Gamma$ is a subgroup of infinite index, show that $\text{cd } \Gamma' < n$. [Hint: For any Γ' -module M' , $H^n(\Gamma', M') \approx H^n(\Gamma, \text{Coind}_{\Gamma'}^{\Gamma} M')$. Now apply 6.9 and exercise 4b of §III.5.] Give an example to show that the hypothesis on $H^n(\Gamma, \mathbb{Z}\Gamma)$ cannot be dropped.
5. In this exercise it will be convenient to think of coefficient modules for homology as being *right* Γ -modules and coefficient modules for cohomology as *left* modules. Recall from §V.3 that there is an evaluation pairing

$$H_i(\Gamma, M) \otimes H^i(\Gamma, N) \rightarrow M \otimes_{\mathbb{Z}\Gamma} N.$$

In particular, taking $N = \mathbb{Z}\Gamma$, we obtain a map

$$(*) \quad H_i(\Gamma, M) \otimes H^i(\Gamma, \mathbb{Z}\Gamma) \rightarrow M \otimes_{\mathbb{Z}\Gamma} \mathbb{Z}\Gamma \approx M.$$

(a) Show that this map is a homomorphism of right Γ -modules, where Γ acts on the domain via its action on $H^i(\Gamma, \mathbb{Z}\Gamma)$. Deduce a map

$$\psi: H_i(\Gamma, M) \rightarrow \text{Hom}_{\Gamma}(H^i(\Gamma, \mathbb{Z}\Gamma), M).$$

[Hint: Use the naturality of the evaluation pairing with respect to module homomorphisms. Alternatively, simply check definitions on the chain level.]

(b) If Γ is of type FP and $n = \text{cd } \Gamma$, show that ψ is an isomorphism in dimension n . Thus

$$(6.10) \quad H_n(\Gamma, M) \approx \text{Hom}_{\Gamma}(D, M) = H^0(\Gamma, \text{Hom}(D, M)),$$

where $D = H^n(\Gamma, \mathbb{Z}\Gamma)$ as in 6.9. [Hint: Consider the natural map $M \otimes_{\mathbb{Z}\Gamma} P \rightarrow \mathcal{H}\text{om}_{\Gamma}(\bar{P}, M)$, where $\bar{P} = \mathcal{H}\text{om}_{\Gamma}(P, \mathbb{Z}\Gamma)$.]

(c) Under the hypotheses of (b), let $z \in H_n(\Gamma, D)$ correspond under 6.10 to $\text{id}_D \in \text{Hom}_{\Gamma}(D, D)$. We call z the *fundamental class* of Γ . By definition, it is characterized by the equation

$$\langle z, u \rangle = u$$

for every $u \in H^n(\Gamma, \mathbb{Z}\Gamma) = D$, where $\langle \quad, \quad \rangle$ denotes the evaluation pairing given by (*) above. Show for any module M that

$$\psi^{-1}: \text{Hom}_{\Gamma}(D, M) \xrightarrow{\cong} H_n(\Gamma, M)$$

is given by $\psi^{-1}(f) = f_*$ for $f \in \text{Hom}_{\Gamma}(D, M)$, where $f_* = H_n(\Gamma, f): H_n(\Gamma, D) \rightarrow H_n(\Gamma, M)$. [Hint: This is true by the definition of z if $M = D$ and $f = \text{id}$; the general case follows from the naturality of ψ^{-1} , cf. exercise 3a of §1.7.] Deduce that the isomorphism

$$H^0(\Gamma, \text{Hom}(D, M)) \xrightarrow{\cong} H_n(\Gamma, M)$$

of 6.10 is given by cap product with z ; more precisely, it is the composite

$$H^0(\Gamma, \text{Hom}(D, M)) \xrightarrow{\cong} H_n(\Gamma, \text{Hom}(D, M) \otimes D) \rightarrow H_n(\Gamma, M),$$

where the second map is induced by the obvious coefficient homomorphism $\text{Hom}(D, M) \otimes D \rightarrow M$. [Hint: Use the description of the cap product $H^0 \otimes H_n \rightarrow H_n$ given in exercise 1 of §V.3.]

(d) Show that the isomorphism

$$\varphi^{-1}: H^n(\Gamma, M) \xrightarrow{\cong} D \otimes_{\mathbb{Z}\Gamma} M$$

of 6.8 is given by evaluation on the fundamental class z . [Hint: This is true by the definition of z if $M = \mathbb{Z}\Gamma$; the general case follows by general nonsense, as in the first proof of 6.8.] Deduce that the isomorphism

$$H^n(\Gamma, M) \xrightarrow{\cong} H_0(\Gamma, D \otimes M)$$

of 6.9 is given by cap product with z . [Hint: Use V.3.10.]

6. (a) Let Γ be of type FP and let D and n be as above. Show that $H_n(\Gamma, D) \neq 0$ and deduce that $\text{cd } \Gamma = \text{hd } \Gamma$, where $\text{hd } \Gamma$, the *homological dimension* of Γ is defined by

$$\text{hd } \Gamma = \sup\{n: H_n(\Gamma, M) \neq 0 \text{ for some } \Gamma\text{-module } M.\}$$

*(b) Give an example of a group Γ (necessarily *not* of type FP) such that $\text{hd } \Gamma < \text{cd } \Gamma$.

*7. If Γ is of type FP and $\text{cd } \Gamma = n$, show that there are spectral sequences

$$E_{pq}^2 = \text{Tor}_p^\Gamma(H^{n-q}(\Gamma, \mathbb{Z}\Gamma), M) \Rightarrow H^{n-(p+q)}(\Gamma, M)$$

and

$$E_2^{pq} = \text{Ext}_p^\Gamma(H^{n-q}(\Gamma, \mathbb{Z}\Gamma), M) \Rightarrow H_{n-(p+q)}(\Gamma, M).$$

By considering the corner $p = q = 0$, recover the isomorphisms $H_n(\Gamma, M) \approx \text{Hom}_\Gamma(D, M)$ and $H^n(\Gamma, M) \approx D \otimes_{\mathbb{Z}\Gamma} M$. [Hint: Use the universal coefficient spectral sequences, cf. Godement [1958], I.5.4.1 and I.5.5.1.]

*8. Prove that the FP and FL properties behave well with respect to extensions, amalgamations, etc. More precisely:

(a) Suppose $1 \rightarrow \Gamma' \rightarrow \Gamma \rightarrow \Gamma'' \rightarrow 1$ is exact. If Γ' and Γ'' are of type FP (resp. FL), then so is Γ .

(b) Let X be an acyclic Γ -complex such that X has only finitely many cells mod Γ . If each isotropy group Γ_σ is of type FP (resp. FL), then so is Γ . In particular, an amalgamation $\Gamma_1 *_A \Gamma_2$ is of type FP (or FL) if Γ_1 , Γ_2 , and A are of type FP (or FL).

[Hint: See Serre [1971] or Bieri [1976].]

7 Topological Interpretation

We have seen (cf. 2.2) that the existence of a finite dimensional $K(\Gamma, 1)$ implies that Γ has finite cohomological dimension. Similarly, 6.3 and 6.4 show that the existence of a finite (resp. finitely dominated) $K(\Gamma, 1)$ implies that Γ is of type FL (resp. FP). The purpose of this section is to consider the converse implications. We also want to give a topological interpretation of the Γ -modules $H^*(\Gamma, \mathbb{Z}\Gamma)$ which arose in §6. We will require a tiny bit of homotopy theory, namely, the Hurewicz theorem (which we quoted in §II.5).

The following theorem is due to Eilenberg–Ganea [1957] and Wall [1965, 1966]:

(7.1) Theorem. *Let Γ be an arbitrary group and let $n = \max\{\text{cd } \Gamma, 3\}$. Then there exists an n -dimensional $K(\Gamma, 1)$ -complex Y . If Γ is finitely presented and of type FL (resp. FP) then Y can be taken to be finite (resp. finitely dominated).*

(We allow here the possibility that $\text{cd } \Gamma = \infty$, in which case the theorem simply asserts the existence of a $K(\Gamma, 1)$ -complex.)

As an immediate consequence of 7.1 we have:

(7.2) Corollary (Eilenberg–Ganea). *If $\text{cd } \Gamma \geq 3$ then $\text{cd } \Gamma = \text{geom dim } \Gamma$.*

Of course we also have $\text{cd } \Gamma = \text{geom dim } \Gamma$ if $\text{cd } \Gamma = 0$ (since Γ is then trivial) or if $\text{cd } \Gamma = 1$ (by the Stallings–Swan theorem which we quoted in Example 2 of §2). In view of Theorem 7.1, then, we always have $\text{cd } \Gamma = \text{geom dim } \Gamma$ except possibly if $\text{cd } \Gamma = 2$ and $\text{geom dim } \Gamma = 3$. It is not known whether this possibility can actually occur.

PROOF OF 7.1. We will construct the skeleta Y^k of the desired Y inductively. To start the induction, let Y^2 be the 2-complex associated to some presentation of Γ (cf. §II.5, Exercise 2); thus $\pi_1 Y^2 \approx \Gamma$. If Γ is finitely presented, Y^2 can be taken to be finite. Note that its universal cover X^2 has $H_i = 0$ for $0 < i < 2$. Now assume inductively that Y^{k-1} has been constructed and that its universal cover X^{k-1} has $H_i = 0$ for $0 < i < k-1$. If Γ is finitely presented and of type FP , assume further that Y^{k-1} is finite. Choose a set of generators (z_α) for the Γ -module $H_{k-1}X^{k-1}$. By the Hurewicz theorem we can find for each α a map $f_\alpha: S^{k-1} \rightarrow X^{k-1}$ which represents z_α , in the sense that $H_{k-1}(f_\alpha): H_{k-1}S^{k-1} \rightarrow H_{k-1}X^{k-1}$ sends a generator of $H_{k-1}S^{k-1}$ to z_α . We now set

$$Y^k = Y^{k-1} \cup \bigcup_{\alpha} e_{\alpha}^k,$$

where the k -cell e_{α}^k is attached to Y^{k-1} via the composite

$$S^{k-1} \xrightarrow{f_{\alpha}} X^{k-1} \rightarrow Y^{k-1}.$$

Letting X^k be the universal cover of Y^k , we must verify that $H_i X^k = 0$ for $0 < i < k$. Note first that we can view X^{k-1} as the $(k-1)$ -skeleton of X^k ; indeed, X^k is obtained from X^{k-1} by attaching k -cells via the maps f_{α} and their transforms under the action of Γ on X^{k-1} . It is clear, then, that $H_i X^k = H_i X^{k-1} = 0$ for $0 < i < k-1$ and that we have an exact sequence

$$H_k(X^k, X^{k-1}) \xrightarrow{\partial} H_{k-1}X^{k-1} \rightarrow H_{k-1}X^k \rightarrow 0.$$

It will therefore suffice to show that ∂ is surjective.

Recall that $H_k(X^k, X^{k-1})$ (which is simply $C_k(X^k)$, the k -th cellular chain group) is a free $\mathbb{Z}\Gamma$ -module with one basis element for each k -cell of Y^k , i.e., for each index α . Explicitly, there is a basis (v_{α}) obtained as follows: if $\chi_{\alpha}: (E^k, S^{k-1}) \rightarrow (X^k, X^{k-1})$ is a characteristic map for the cell attached via f_{α} , then $v_{\alpha} \in H_k(X^k, X^{k-1})$ is defined to be the image under

$$H_k(\chi_{\alpha}): H_k(E^k, S^{k-1}) \rightarrow H_k(X^k, X^{k-1})$$

of a generator of $H_k(E^k, S^{k-1}) = \mathbb{Z}$. In view of the diagram

$$\begin{array}{ccc} H_k(E^k, S^{k-1}) & \xrightarrow{\partial} & H_{k-1}S^{k-1} \\ \downarrow H_k(\chi_{\alpha}) & & \downarrow H_{k-1}(f_{\alpha}) \\ H_k(X^k, X^{k-1}) & \xrightarrow{\partial} & H_{k-1}X^{k-1}, \end{array}$$

it follows that $\partial v_{\alpha} = z_{\alpha}$ (assuming that the generators of $H_k(E^k, S^{k-1})$ and $H_{k-1}S^{k-1}$ have been chosen compatibly), so that ∂ is indeed surjective.

(Note for future reference that if $H_{k-1}X^{k-1}$ happens to be a *free* $\mathbb{Z}\Gamma$ -module with basis (z_α) , then $\partial: H_k(X^k, X^{k-1}) \rightarrow H_{k-1}X^{k-1}$ is an isomorphism. It then follows from the long exact homology sequence of the pair (X^k, X^{k-1}) that $H_iX^k = 0$ for all $i > 0$, so that Y^k is a $K(\Gamma, 1)$.)

To complete the inductive step, we must show that Y^k can be taken to be finite if Γ is finitely presented and of type *FP*, i.e., we must show in this case that $H_{k-1}X^{k-1}$ is a finitely generated Γ -module. To see this, we need only note that the cellular chain complex

$$C_{k-1} \rightarrow \cdots \rightarrow C_0 \rightarrow \mathbb{Z} \rightarrow 0$$

of X^{k-1} is a partial free resolution of *finite type*, since Y^{k-1} was assumed to be finite. Therefore $H_{k-1} = \ker\{C_{k-1} \rightarrow C_{k-2}\}$ is finitely generated by 4.3 since Γ is of type *FP*.

If $n = \infty$, we now continue this inductive process indefinitely, and $Y = \bigcup_k Y^k$ is the desired $K(\Gamma, 1)$. If $n < \infty$, consider X^{n-1} . (This makes sense because $n - 1 \geq 2$.) Its cellular chain complex

$$C_{n-1} \rightarrow \cdots \rightarrow C_0 \rightarrow \mathbb{Z} \rightarrow 0$$

is a partial free resolution of length $n - 1$. Hence 2.1 implies that $H_{n-1}X^{n-1}$ is a projective $\mathbb{Z}\Gamma$ -module. By the Eilenberg trick (2.7) there is a free module F such that $H_{n-1}X^{n-1} \oplus F$ is free. We now replace Y^{n-1} by $\bar{Y}^{n-1} = Y^{n-1} \vee S^{n-1} \vee S^{n-1} \vee \cdots$, where there is one copy of S^{n-1} for each basis element of F . The effect of this on $C(X^{n-1})$ is simply to add F to C_{n-1} , with $\partial|_F = 0$. The universal cover $\bar{X}^{n-1}$ now has H_{n-1} free. We may therefore attach n -cells e_α^n to $\bar{Y}^{n-1}$ corresponding to basis elements z_α of $H_{n-1}\bar{X}^{n-1}$; as remarked above, the resulting $\bar{Y}^n = \bar{Y}^{n-1} \cup \bigcup e_\alpha^n$ will then be an n -dimensional $K(\Gamma, 1)$.

Suppose now that Γ is finitely presented and of type *FL*. Then Y^{n-1} is finite and the projective $H_{n-1}X^{n-1}$ is finitely generated. We know from 6.5 that $H_{n-1}X^{n-1}$ is stably free, so that there is a free module F of *finite rank* such that $H_{n-1}X^{n-1} \oplus F$ is free of finite rank. We now proceed as in the previous paragraph, and the resulting $\bar{Y}^n$ will be a finite $K(\Gamma, 1)$.

Finally, suppose that Γ is finitely presented but only of type *FP* instead of *FL*. On the one hand, the general inductive step above gives us a finite complex

$$Y^n = Y^{n-1} \cup e^n \cup \cdots \cup e^n$$

whose universal cover has $H_i = 0$ for $0 < i < n$. Hence $\pi_i Y^n \approx \pi_i X^n = 0$ for $1 < i < n$. On the other hand, we know that there is a $K(\Gamma, 1)$ of the form

$$\bar{Y}^n = Y^{n-1} \vee S^{n-1} \vee \cdots \cup e^n \cup \cdots,$$

so that $\pi_i \bar{Y}^n = 0$ for all $i > 1$. I claim that Y^n dominates $\bar{Y}^n$. Indeed, the required maps

$$\bar{Y}^n \xrightarrow{i} Y^n \xrightarrow{r} \bar{Y}^n$$

with $ri \simeq \text{id}$ are easily constructed as follows: i and r are both defined to be the identity on the common subcomplex Y^{n-1} , and they are extended arbitrarily to the cells that were attached. (These extensions exist trivially for each S^{n-1} that was wedged onto Y^{n-1} in forming $\bar{Y}^n$, and they exist for each e^n because $\pi_{n-1}Y^n = 0$ and $\pi_{n-1}\bar{Y}^n = 0$.) Finally, the homotopy $ri \simeq \text{id}$ is defined to be the constant homotopy on Y^{n-1} and is extended to all of $\bar{Y}^n$ by means of the vanishing of $\pi_{n-1}\bar{Y}^n$ and $\pi_n\bar{Y}^n$. $\square$

For future reference we mention the following refinement of 7.1:

(7.3) Addendum. *The complex Y in 7.1 can be taken to be a simplicial complex.*

PROOF. Assume inductively that Y^{k-1} is simplicial, where the notation is that of the proof above. By the simplicial approximation theorem, we may then take each attaching map $f_\alpha: S^{k-1} \rightarrow Y^{k-1}$ to be simplicial relative to some triangulation of S^{k-1} . The resulting space Y^k is then triangulable by Whitehead [1949], §9, Lemma 2. $\square$

Finally, we give a topological interpretation of the right Γ -modules $H^*(\Gamma, \mathbb{Z}\Gamma)$, assuming that Γ is finitely presented and of type FL . We will need the following observation:

(7.4) Lemma. *Let Γ be a group and M a left Γ -module. Let $\text{Hom}_\Gamma(M, \mathbb{Z}) \subseteq \text{Hom}(M, \mathbb{Z})$ consist of all abelian group homomorphisms $f: M \rightarrow \mathbb{Z}$ such that, for every $m \in M$, $f(\gamma m) = 0$ for all but finitely many $\gamma \in \Gamma$. Then there is a natural isomorphism*

$$\text{Hom}_\Gamma(M, \mathbb{Z}\Gamma) \approx \text{Hom}_\Gamma(M, \mathbb{Z}).$$

Moreover, this is an isomorphism of right Γ -modules, where Γ acts on $\text{Hom}_\Gamma(M, \mathbb{Z}\Gamma)$ via its right action on $\mathbb{Z}\Gamma$ and Γ acts on $\text{Hom}_\Gamma(M, \mathbb{Z})$ via its left action on M (i.e., $(f\gamma)(m) = f(\gamma m)$ for $f \in \text{Hom}_\Gamma(M, \mathbb{Z})$, $\gamma \in \Gamma$, $m \in M$).

PROOF. A $\mathbb{Z}$ -module map $F: M \rightarrow \mathbb{Z}\Gamma$ has the form

$$F(m) = \sum_{\gamma \in \Gamma} f_\gamma(m)\gamma,$$

where $f_\gamma: M \rightarrow \mathbb{Z}$ and, for each $m \in M$, $f_\gamma(m) = 0$ for almost all $\gamma \in \Gamma$. One checks that such an F is a Γ -module homomorphism if and only if $f_\gamma(m) = f_1(\gamma^{-1}m)$ for all $\gamma \in \Gamma$. We therefore have a map $\text{Hom}_\Gamma(M, \mathbb{Z}\Gamma) \rightarrow \text{Hom}_\Gamma(M, \mathbb{Z})$ given by $F \mapsto f_1$, and this map is an isomorphism with inverse

$$f \mapsto \left\{ m \mapsto \sum_{\gamma \in \Gamma} f(\gamma^{-1}m)\gamma \right\}.$$

The reader can easily verify that this isomorphism is natural and compatible with the right Γ -actions. $\square$

Suppose now that Γ is finitely presented and of type *FL*. By 7.1 there is a contractible, free Γ -complex X with X/Γ finite. Then $C(X)$ is a finite free resolution of $\mathbb{Z}$ over $\mathbb{Z}\Gamma$, and $H^*(\Gamma, \mathbb{Z}\Gamma)$ is the cohomology of

$$\mathcal{H}om_{\Gamma}(C(X), \mathbb{Z}\Gamma).$$

In view of the lemma, we have

$$\mathcal{H}om_{\Gamma}(C(X), \mathbb{Z}\Gamma) \approx \mathcal{H}om_c(C(X), \mathbb{Z}) \subseteq \mathcal{H}om(C(X), \mathbb{Z}),$$

this isomorphism being compatible with the right Γ -action and the co-boundary operators (the latter because of the naturality assertion in 7.4).

Recall that $C(X)$ has a $\mathbb{Z}$ -basis with one element for each cell σ of X . These basis elements are freely permuted by Γ and fall into finitely many orbits. It follows easily that $\mathcal{H}om_c(C(X), \mathbb{Z})$ consists of those cochains $f \in \mathcal{H}om(C(X), \mathbb{Z})$ such that $f(\sigma) = 0$ for all but finitely many cells σ . The cohomology of this complex is called the cohomology of X with *compact supports*, and is denoted $H_c^*(X; \mathbb{Z})$. We have now established:

(7.5) Proposition. *If X is a contractible, free Γ -complex with compact quotient X/Γ , then there is an isomorphism*

$$H^*(\Gamma, \mathbb{Z}\Gamma) \approx H_c^*(X; \mathbb{Z})$$

of right Γ -modules, where the right action of Γ on $H_c^(X; \mathbb{Z})$ is induced by the left action of Γ on X .*

In view of 6.7, this yields:

(7.6) Corollary. *If X is as in 7.5, then*

$$\text{cd } \Gamma = \max\{n : H_c^n(X; \mathbb{Z}) \neq 0\}.$$

EXERCISES

1. If Γ is a countable group, show that there exists a countable $K(\Gamma, 1)$ -complex.
2. Give a topological interpretation of the FP_n conditions ($3 \leq n \leq \infty$) assuming Γ is finitely presented.
3. Suppose $\text{cd } \Gamma = 2$. Show that there exists a 2-dimensional, acyclic, free Γ -complex. [Hint: Write $\Gamma = F/R$, where F is free and $R \triangleleft F$. Let Y^1 be a 1-complex with $\pi_1 = F$, let X^1 be the covering space corresponding to R , and argue as in the proof of 7.1.] Thus for any Γ we can characterize $\text{cd } \Gamma$ topologically as the minimal dimension of a free, acyclic Γ -complex.
- *4. Prove the following generalization of 7.5: Let X be a contractible Γ -complex with finite isotropy groups Γ_σ and with only finitely many cells mod Γ . Then $H^*(\Gamma, \mathbb{Z}\Gamma) \approx H_c^*(X; \mathbb{Z})$. [Hint: First show, by a spectral sequence argument for instance, that $H^*(\Gamma, \mathbb{Z}\Gamma)$ can be computed from $\mathcal{H}om_{\Gamma}(C(X), \mathbb{Z}\Gamma)$.]

8 Further Topological Results

The purpose of this section is to see what finiteness properties of Γ can be deduced if we are given a $K(\Gamma, 1)$ which is a *manifold*. We will see many examples of this situation in the next section. The proofs to be given in this section require more background in topology than we have assumed elsewhere in this book. The reader without this background, however, can still read and understand the *statements* of all the results.

We begin by noting the analogues of 2.2 and 6.3 for $K(\Gamma, 1)$ -manifolds:

(8.1) Proposition. *Suppose Y is a d -dimensional $K(\Gamma, 1)$ -manifold (possibly with boundary).*

- (a) *$\text{cd } \Gamma \leq d$, with equality if and only if Y is closed (i.e., compact and without boundary).*
- (b) *If Y is compact then Γ is of type FL .*

PROOF. (a) If Y is smooth, as it will be in all of our examples, then there are several ways to show that Y has the homotopy type of a CW -complex Y' of dimension $\leq d$ (so that $\text{cd } \Gamma \leq d$ by 2.2): one can use Whitehead's triangulation theorem (cf. Munkres [1966], Ch. II), or Morse theory (cf. Milnor [1963]), or the theory of nerves of coverings (cf. Weil [1952]). An alternative, which works even if Y is not smooth, is to deduce from Poincaré duality with local coefficients that $H^i(\Gamma, M) = H^i(Y; M) = 0$ for $i > d$ and all Γ -modules M . This proves the first part of (a). If Y is closed, then we have $H^d(\Gamma, \mathbb{Z}_2) \approx H^d(Y; \mathbb{Z}_2) = \mathbb{Z}_2 \neq 0$, so $\text{cd } \Gamma = d$. If Y is not closed, then one can deduce from Poincaré duality with local coefficients that $H^d(\Gamma, M) = H^d(Y; M) = 0$ for all Γ -modules M , so that $\text{cd } \Gamma < d$; we omit the details since we will not make serious use of this strict inequality. [Alternatively, if Y is assumed to be triangulated, then there is a geometric proof that $\text{cd } \Gamma < d$: one shows that Y admits a deformation retraction onto a subcomplex of dimension $< d$. We will describe such a deformation retraction explicitly in an interesting example in §9, Example 3.]

(b) Suppose Y is compact. In the smooth case, the complex Y' in the proof of (a) can be taken to be finite, so Γ is of type FL by 6.3. In the general case it is still true, but considerably harder to prove, that Y has the homotopy type of a finite complex (cf. Kirby-Siebenmann [1969] or West [1977]), so (b) is proved. [It is worth noting here that if we are content to prove that Γ is of type FP , then there is a quite elementary proof: we need only embed Y in Euclidean space and note that it is a retract of a compact polyhedral neighborhood (cf. Dold [1972], proof of V.4.11); thus Γ is of type FP by 6.4.] $\square$

Next we wish to reinterpret $H^*(\Gamma, \mathbb{Z}\Gamma)$ (cf. 7.5) in case there is a compact $K(\Gamma, 1)$ -manifold Y . Let X be the universal cover of Y . Since X is simply-connected, it is certainly orientable, and we denote by Ω its "orientation

module." Thus Ω is an infinite cyclic group whose two generators correspond to the two orientations of X . The action of Γ on X induces an action of Γ on Ω , with an element $\gamma \in \Gamma$ acting as ± 1 according as the action of γ on X is orientation-preserving or orientation-reversing. Note that Γ acts trivially on Ω if and only if Y is orientable. Finally, we make the convention that the reduced homology of a space Z , denoted $\tilde{H}_*(Z)$, is the homology of the augmented chain complex of Z . In particular, $\tilde{H}_{-1}(\emptyset) = \mathbb{Z}$. We can now state:

(8.2) Proposition. *Let Y be a compact d -dimensional $K(\Gamma, 1)$ -manifold (possibly with boundary). Let X be its universal cover and let Ω be the corresponding orientation module. Then there are Γ -module isomorphisms*

$$H^i(\Gamma, \mathbb{Z}\Gamma) \approx \tilde{H}_{d-i-1}(\partial X) \otimes \Omega$$

for all i . In particular, if Y is a closed manifold, then

$$H^i(\Gamma, \mathbb{Z}\Gamma) \approx \begin{cases} 0 & i \neq d \\ \Omega & i = d. \end{cases}$$

(8.3) Corollary. *Under the hypotheses of 8.2, there exists at least one integer k such that $\tilde{H}_k(\partial X) \neq 0$. Moreover, letting*

$$l = 1 + \min\{k: \tilde{H}_k(\partial X) \neq 0\},$$

we have

$$\text{cd } \Gamma = d - l.$$

The corollary is immediate from 8.2 and 6.7. Note that $l > 0$ if $\partial Y \neq \emptyset$. Thus 8.3 makes more precise the inequality $\text{cd } \Gamma < d$ of 8.1a in this case.

PROOF OF 8.2. Since Y is compact and has the homotopy type of a finite complex, we have

$$H^i(\Gamma, \mathbb{Z}\Gamma) \approx H_c^i(X)$$

by 7.5. On the other hand, there is a Poincaré–Lefschetz duality isomorphism

$$H_c^i(X) \approx H_{d-i}(X, \partial X).$$

This, however, is not canonical; it depends on a choice of orientation of X . In particular, it commutes or anti-commutes with the action of an element $\gamma \in \Gamma$ according as γ preserves or reverses the orientation of X . Consequently, we have a Γ -module isomorphism

$$H_c^i(X) \approx H_{d-i}(X, \partial X) \otimes \Omega.$$

Finally, since $\tilde{H}_*(X) = 0$,

$$H_{d-i}(X, \partial X) \approx \tilde{H}_{d-i-1}(\partial X).$$

The proposition follows at once. □

Remark. The reader who is uncomfortable with Poincaré–Lefschetz duality for non-compact manifolds with boundary might prefer the following alternative proof, which uses the duality theorem only for the compact manifold Y (but with local coefficients) and which does not use the fact that Y has the homotopy type of a finite complex:

Regarding $\mathbb{Z}\Gamma$ and Ω as local coefficient systems on Y , we have

$$H^i(\Gamma, \mathbb{Z}\Gamma) \approx H^i(Y; \mathbb{Z}\Gamma) \approx H_{d-i}(Y, \partial Y; \mathbb{Z}\Gamma \otimes \Omega).$$

Now $\mathbb{Z}\Gamma \otimes \Omega$, with diagonal Γ -action, is an induced module (cf. III.5.7), and homology with coefficients in an induced module is easily seen to be ordinary homology of the universal cover. One deduces

$$H_{d-i}(Y, \partial Y; \mathbb{Z}\Gamma \otimes \Omega) \approx H_{d-i}(X, \partial X; \Omega) \approx H_{d-i}(X, \partial X) \otimes \Omega,$$

and the result now follows as above.

EXERCISE

Let Γ be a group such that there exists a closed $K(\Gamma, 1)$ -manifold. If $\Gamma' \subset \Gamma$ is a subgroup of infinite index, deduce from 8.1a that $\text{cd } \Gamma' < \text{cd } \Gamma$. [Hint: If $p: \tilde{Y} \rightarrow Y$ is a covering map of infinite degree, then $\tilde{Y}$ cannot be compact.] See Exercise 6 of §10 below for a generalization of this result.

9 Further Examples

We saw a few examples in §2 of groups Γ with $\text{cd } \Gamma < \infty$, and we noted in §6 that some of those examples were of type *FL*. We now want to introduce some additional families of examples, the most interesting of which are the “arithmetic” groups. The study of such examples leads, as we will see in the next chapter, to some remarkable connections between group cohomology theory and number theory. Unfortunately, the assertions which we will make about the examples in this section are considerably less elementary than the corresponding assertions concerning our previous examples, and we will not be able to give the proofs. The reader is therefore advised to casually read through this section, taking on faith a number of deep results which we will have to state without proof.

1. Let Γ be a classical knot group (i.e., the fundamental group of the complement of a non-trivial knot $K \hookrightarrow S^3$). Then Γ is of type *FL* and $\text{cd } \Gamma = 2$. To see this, let $Y = S^3 - T$, where T is an open tubular neighborhood of K . Then Y is a compact 3-manifold whose boundary is a torus, and a deep theorem of Papakyriakopoulos [1957] says that Y is a $K(\Gamma, 1)$. It now follows from 8.1 that Γ is of type *FL*. To calculate $\text{cd } \Gamma$, we use another result from knot theory, namely, that $\pi_1(\partial Y)$ injects into $\pi_1 Y$ as a subgroup

of infinite index. If X is the universal cover of Y , it follows that ∂X is the disjoint union of countably many copies of the universal cover $\mathbb{R}^2$ of ∂Y . Thus the integer l in 8.3 is 1, and we have $\text{cd } \Gamma = 3 - 1 = 2$, as claimed. Furthermore, we can use 8.2 to compute $H^*(\Gamma, \mathbb{Z}\Gamma)$. In particular, we see that $H^i(\Gamma, \mathbb{Z}\Gamma) = 0$ for $i \neq 2$ and that $H^2(\Gamma, \mathbb{Z}\Gamma)$ is a free abelian group of countable rank.

2. Let Γ be the $n \times n$ strict upper triangular group over $\mathbb{Z}$. We saw in Example 5 of §II.4 how to construct a $K(\Gamma, 1)$ -manifold (without boundary). Namely, take $Y = \Gamma \backslash G$ where G is the $n \times n$ strict upper triangular group over $\mathbb{R}$. It is easy to see that Y is compact (see the exercise below), so 8.1 implies that Γ is of type FL and that

$$\text{cd } \Gamma = \dim Y = n(n-1)/2.$$

(In particular, if $n = 3$, we recover the result of Example 6 of §2.) Moreover, 8.2 shows that

$$H^i(\Gamma, \mathbb{Z}\Gamma) = \begin{cases} 0 & i \neq \frac{n(n-1)}{2} \\ \mathbb{Z} & i = \frac{n(n-1)}{2}, \end{cases}$$

where Γ acts trivially on $\mathbb{Z}$. More generally, one can prove analogous results for an arbitrary finitely generated, torsion-free, nilpotent group Γ . For Malcev [1949] proved that Γ can be embedded as a discrete subgroup with compact quotient in a nilpotent Lie group G which is homeomorphic to Euclidean space of dimension $d = \text{rank } \Gamma$. [One can view G as a “tensor product” $\mathbb{R} \otimes \Gamma$, cf. Bourbaki, LIE II, pp. 82–83, and LIE III, pp. 283–284.]

3. Consider now the group $SL_n(\mathbb{Z})$, $n \geq 2$. This group has torsion, hence $\text{cd}(SL_n(\mathbb{Z})) = \infty$. We know, however, that it has torsion-free subgroups Γ of finite index, cf. Exercise 3 of §II.4. The intersection of Γ with the strict upper triangular group U has finite index in U and hence has $\text{cd} = n(n-1)/2$ by Example 2 and Prop. 2.4a. Thus

$$(9.1) \quad \text{cd } \Gamma \geq \frac{n(n-1)}{2}.$$

We will now outline a proof, based on the reduction theory of quadratic forms, that Γ is of type FL and that equality holds in 9.1. A different proof of these results will be discussed in Example 5 below.

Let X be the space which we called X_0 in Example 7 of §II.4; thus X is the space of positive definite quadratic forms on $\mathbb{R}^n$, modulo multiplication by positive scalars. Recall that X is a contractible manifold (without boundary) of dimension $n(n+1)/2 - 1$, that $SL_n(\mathbb{R})$ (hence also Γ) acts on X , and that X/Γ is a $K(\Gamma, 1)$. Recall also that if $n = 2$ then we can identify

X with the upper half plane (or, equivalently, with the open unit disk), with $SL_n(\mathbb{R})$ acting by linear fractional transformations.

In view of 8.1a we have

$$\text{cd } \Gamma \leq \frac{n(n+1)}{2} - 1.$$

Moreover, one can show that X/Γ is non-compact, so that 8.1a implies that strict inequality holds above. But we can do substantially better than this by giving a direct geometric construction instead of relying on the generalities given in 8.1. Namely, we will show that X/Γ admits a deformation retraction onto a subspace which is a CW -complex of dimension $n(n-1)/2$. This will show that $\text{cd } \Gamma \leq n(n-1)/2$ and hence will prove our claim that equality holds in 9.1.

We begin by describing how this is done for $n = 2$, using the open unit disk as our model for X . There is a tiling of X by “ideal hyperbolic triangles,” which is compatible with the action of $SL_2(\mathbb{Z})$, and which is well-known in the theory of modular forms (see, for instance, Lehner [1964]). It is obtained by starting with a single ideal triangle (i.e., a hyperbolic triangle with vertices on the unit circle) and generating further triangles by successive reflections across the sides. This is illustrated in Fig. 9.2a, where alternate triangles of the tiling are shaded. The vertices of the triangles of the tiling are called cusps, and we denote by X^* the space obtained from X by adjoining the cusps; it can be viewed as a simplicial complex with a simplicial $SL_2(\mathbb{Z})$ -action. [Note: We give X^* the usual simplicial topology, rather than the topology it inherits as a subset of the plane. In particular, the set ∂X^* of cusps is a discrete set in

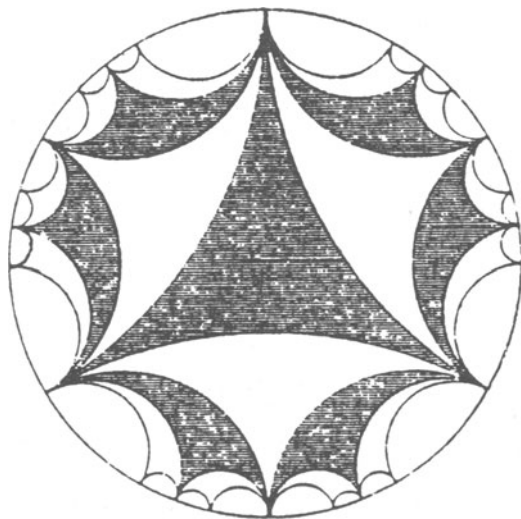


Figure 9.2a

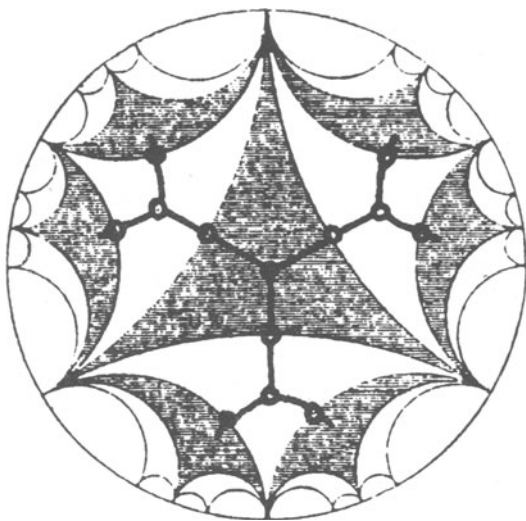


Figure 9.2b

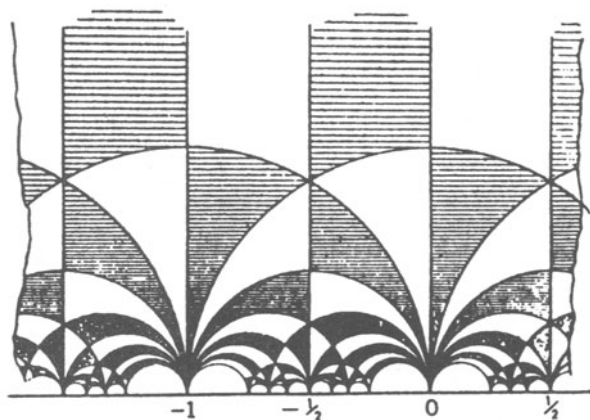


Figure 9.2c

(Figures 9.2a and 9.2c are reproduced from F. Klein and R. Fricke, *Vorlesungen über die Theorie der elliptischen Modulfunktionen*, Band I, B. G. Teubner, Leipzig, 1890, by permission of the publisher.)

the simplicial topology. It can be shown, however, that the simplicial topology agrees with the usual topology on the open subspace X of X^* .]

Now let T be the simplicial complement of ∂X^* in the barycentric subdivision K of X^* , i.e., T is the largest subcomplex of K disjoint from ∂X^* . (Explicitly, T consists of all simplices of K none of whose vertices are in ∂X^* .) The subcomplex T is shown in Fig. 9.2b. See also Fig. 9.2c, where the

entire barycentric subdivision K is shown in the upper half plane model for X . The reader is invited to trace out the subcomplex T in this picture. [Note: The vertices of the large triangle in Fig. 9.2a correspond to the points $-1, 0, \infty$ in Fig. 9.2c; the barycenter of this triangle corresponds to the point $(-1 + i\sqrt{3})/2$ in Fig. 9.2c.]

I claim that this simplicial complement T is, in a canonical way, a deformation retract of the geometric complement $X = X^* - \partial X^*$. Indeed, one deforms X to T by pushing away from ∂X^* along straight lines (in the hyperbolic or simplicial sense). More precisely, any $x \in X$ lies in a closed 2-simplex σ of K with one vertex v in ∂X^* and the opposite face τ in T . Since $x \neq v$, there is a well-defined ray from v to x , and the deformation moves x along this ray away from v until it hits τ .

Note that T and the deformation are described purely in terms of the simplicial structure on X^* , so they are compatible with the action of $SL_2(\mathbb{Z})$ and its subgroup Γ . It follows that T/Γ is a deformation retract of X/Γ , and it has dimension $1 = 2(2 - 1)/2$, as required.

Before proceeding to the general case, we make some remarks about the construction above. Let K be a simplicial complex. A subcomplex L is said to be *full* if every simplex of K having all of its vertices in L is itself in L . For example, the subcomplex ∂X^* above is not full in X^* , but it is full in the barycentric subdivision of X^* . (More generally, if K is an arbitrary complex and L an arbitrary subcomplex, then the barycentric subdivision of L is a full subcomplex of the barycentric subdivision of K .) Let T be the simplicial complement of L in K ; this is defined, as above, to be the subcomplex consisting of all simplices of K which have no vertices in L . If L is full in K , then every simplex of K which is not in L has a non-empty face in T such that the opposite face (possibly empty) is in L . It follows easily that the geometric complement $K - L$ admits a deformation retraction onto the simplicial complement T ; one simply pushes along straight lines from the L -face of any simplex to the T -face.

What we did above, then, can now be explained as follows: We first passed to the barycentric subdivision so that ∂X^* would become full, and we then had a deformation retraction of the geometric complement of ∂X^* onto the simplicial complement.

The generalization to $SL_n(\mathbb{Z})$ for arbitrary n is based on a theory due to Voronoi [1907]. Voronoi constructs an enlargement X^* of X , obtained by adjoining certain positive *semi*-definite quadratic forms, namely, those whose nullspace admits a basis consisting of vectors in $\mathbb{Q}^n$. He gives X^* an explicit decomposition into convex cells which are permuted by the action of $SL_n(\mathbb{Z})$. The subspace $\partial X^* = X^* - X$ is a subcomplex. The cells of X are not necessarily simplices, but X^* admits a barycentric subdivision K which is simplicial and which inherits a simplicial action of $SL_n(\mathbb{Z})$. It follows, as above, that $X = X^* - \partial X^*$ admits a deformation retraction (compatible with the $SL_n(\mathbb{Z})$ -action) onto the simplicial complement T of ∂X^* in K . One sees by looking at Voronoi's construction that ∂X^* contains the entire

$(n - 2)$ -skeleton of X^* , and it follows easily that T has codimension $\geq n - 1$. Thus

$$\dim T/\Gamma = \dim T \leq \frac{n(n+1)}{2} - 1 - (n-1) = \frac{n(n-1)}{2},$$

as required. Finally, Voronoi proves in addition that X^* has only finitely many cells mod the $SL_n(\mathbb{Z})$ -action, so T/Γ is in fact a finite complex and Γ is of type FL .

Remark. In case $n = 2$, the subcomplex T is a tree, being a 1-dimensional deformation retract of the contractible space X . This tree was first introduced by Serre [1977a], I.4.2, who used it to give an easy proof of the classical theorem expressing $SL_2(\mathbb{Z})$ as an amalgamation $\mathbb{Z}_4 *_{\mathbb{Z}_2} \mathbb{Z}_6$. The existence of an analogue of T for $n > 2$ (i.e., a Γ -invariant, contractible subcomplex of X of dimension $n(n-1)/2$ with T/Γ compact) was proved by Soulé [1978] for $n = 3$ and by Ash [1977] in general. Ash, in fact, proved an analogous result for a more general class of groups, using a generalization of Voronoi's theory. The cohomological dimension of such groups had previously been computed by Borel and Serre [1974], using different methods which we will describe in Example 5 below.

The groups in Examples 2 and 3 are examples of “arithmetic” groups. Before looking at general arithmetic groups, let's see what can be said about arbitrary discrete subgroups of Lie groups.

4. Let G be a Lie group with only finitely many connected components, let K be a maximal compact subgroup, and let $d = \dim G - \dim K$. If Γ is any torsion-free discrete subgroup of G , I claim that

$$\text{cd } \Gamma \leq d,$$

with equality if and only if Γ is *co-compact* in G , i.e., if and only if G/Γ is compact. Moreover, in the co-compact case Γ is of type FL and one has $H^i(\Gamma, \mathbb{Z}\Gamma) = 0$ for $i \neq d$ and $H^d(\Gamma, \mathbb{Z}\Gamma) = \mathbb{Z}$ (possibly with non-trivial Γ -action). To prove these assertions we need only recall (cf. §II.4, Example 8) that $\Gamma \backslash G/K$ is a $K(\Gamma, 1)$ -manifold (without boundary) of dimension d ; moreover, $\Gamma \backslash G/K$ is compact if and only if Γ is co-compact. The assertions now follow at once from 8.1 and 8.2.

5. We now specialize Example 4 to the “arithmetic” case. We will confine ourselves to a brief outline of this theory, referring to the survey paper of Serre [1979] for more details and further references. Let G be a linear algebraic group defined over $\mathbb{Q}$. In concrete terms, this simply means that G is a subgroup of some general linear group GL_n , defined by polynomial equations (with rational coefficients) in the n^2 matrix entries. For example, G could be the group SL_n (defined by the single equation $\det(a_{ij}) = 1$) or the

strict upper triangular group (defined by the equations $a_{ii} = 1$, $a_{ij} = 0$ for $i > j$). The real matrices satisfying the given polynomial equations then form a Lie group $G(\mathbb{R})$ (e.g., $SL_n(\mathbb{R})$), which is known to have only finitely many components; and the integral matrices satisfying the equations form a discrete subgroup $G(\mathbb{Z}) \subseteq G(\mathbb{R})$ (e.g., $SL_n(\mathbb{Z}) \subset SL_n(\mathbb{R})$). The group $G(\mathbb{Z})$ is said to be an *arithmetic group*. More generally, if $\Gamma \subseteq G(\mathbb{Q})$ is a subgroup *commensurable* with $G(\mathbb{Z})$, i.e., such that $\Gamma \cap G(\mathbb{Z})$ is of finite index in both Γ and $G(\mathbb{Z})$, then Γ is said to be an *arithmetic subgroup* of $G(\mathbb{Q})$. In particular, any subgroup of $G(\mathbb{Z})$ of finite index is arithmetic.

Suppose now that Γ is a *torsion-free* arithmetic group. As in Example 4, we have a contractible manifold $X = G(\mathbb{R})/K$ on which Γ acts freely, so that X/Γ is a $K(\Gamma, 1)$ -manifold. It turns out, however, that Γ is usually *not* co-compact in $G(\mathbb{R})$, and hence that X/Γ is non-compact. In fact, there is a “ $\mathbb{Q}$ -rank” $l \geq 0$ attached to the algebraic group G , and Γ is co-compact in $G(\mathbb{R})$ if and only if $l = 0$.

[Here is the precise definition of l , for the benefit of the reader familiar with algebraic groups: l is the rank of a maximal $\mathbb{Q}$ -split torus in G/RG , where RG is the radical of G . If G is the strict upper triangular group, for example, then $RG = G$, so $l = 0$ and Γ is co-compact, as we saw in Example 2. If $G = SL_n$ ($n \geq 2$), on the other hand, then $RG = \{1\}$ and a maximal split torus in $G/RG = G$ is given by the diagonal matrices; thus $l = n - 1 > 0$ and Γ is not co-compact, as we saw in Example 3.]

In spite of this failure of Γ to be co-compact, however, Borel and Serre [1974] show that Γ is of type FL and compute $H^*(\Gamma, \mathbb{Z}\Gamma)$. Their method is to replace X by a contractible manifold $\bar{X}$ with boundary (with X as its interior if G is semi-simple) on which $G(\mathbb{Q})$ operates. They show that the action of Γ is free and proper and that the quotient $\bar{X}/\Gamma$ is compact. Thus $\bar{X}/\Gamma$ is a compact $K(\Gamma, 1)$ -manifold, so that Γ is of type FL by 8.1b. Moreover, the construction of $\bar{X}$ is explicit enough that Borel and Serre are able to identify the homotopy type of the boundary: namely, $\partial\bar{X}$ has the homotopy type of a countable bouquet of $(l - 1)$ -spheres, where l is the $\mathbb{Q}$ -rank mentioned above. [If $l = 0$, this simply means that $\partial\bar{X} = \emptyset$.] Letting $d = \dim \bar{X}$, we conclude from 8.2 that $H^i(\Gamma, \mathbb{Z}\Gamma) = 0$ for $i \neq d - l$, and that $H^{d-l}(\Gamma, \mathbb{Z}\Gamma)$ is a free abelian group of countable rank if $l > 0$ and of rank 1 if $l = 0$. In particular,

$$\text{cd } \Gamma = d - l.$$

If $G = SL_n$, for example, then $d = n(n + 1)/2 - 1$ and $l = n - 1$, so $\text{cd } \Gamma = d - l = n(n - 1)/2$, as we saw in Example 3.

6. Let S be a finite set of prime numbers and let $\mathbb{Z}_S \subset \mathbb{Q}$ be the localization of $\mathbb{Z}$ obtained by inverting the elements of S . If G is as in example 5, then a subgroup of $G(\mathbb{Q})$ is called *S-arithmetic* if it is commensurable with $G(\mathbb{Z}_S)$. Borel and Serre [1976] have shown that the results of example 5 extend to torsion-free *S-arithmetic* groups, provided G is reductive.

7. Serre [1971] noted the following remarkable consequence of example 6: If Γ is any finitely generated torsion-free subgroup of $GL_n(\mathbb{Q})$, then $\text{cd } \Gamma < \infty$. [Sketch of proof: Note that $\Gamma \subseteq GL_n(\mathbb{Z}_S)$ for some S , and apply example 6 to the S -arithmetic group $GL_n(\mathbb{Z}_S)$ to conclude that any torsion-free subgroup of the latter has finite cohomological dimension.]

8. The result of example 7 is no longer true if $\mathbb{Q}$ is replaced by a field K of transcendence degree ≥ 1 over $\mathbb{Q}$. For one can find finitely generated torsion-free subgroups of $GL_n(K)$ which contain unipotent subgroups that are free abelian of infinite rank. [A subgroup U of $GL_n(K)$ is *unipotent* if every element of U has all n of its eigenvalues equal to 1. According to a famous theorem of Kolchin, this holds iff U is conjugate to a subgroup of the strict upper triangular group.] But Alperin and Shalen [1982] have shown that this is essentially the only type of counter-example. More precisely, they show that if Γ is a finitely generated torsion-free subgroup of $GL_n(K)$, where K is an arbitrary field of characteristic 0, then $\text{cd } \Gamma < \infty$ iff there is a finite upper bound on the Hirsch ranks of the finitely generated unipotent subgroups of Γ .

Remark. The reader may have noticed that all of the examples of groups of type FL which we gave in this section have the following remarkable property: There is an integer n (necessarily equal to $\text{cd } \Gamma$) such that $H^i(\Gamma, \mathbb{Z}\Gamma) = 0$ for $i \neq n$ and $H^n(\Gamma, \mathbb{Z}\Gamma)$ is free abelian. The significance of this, as we will see in the next section, is that it implies that these groups satisfy a *duality* relation, somewhat analogous to Poincaré duality for manifolds.

EXERCISE

Let G and Γ be as in Example 2. Let $C = \{(a_{ij}) \in G : |a_{ij}| \leq 1 \text{ for all } i, j\}$. Show that $G = \Gamma \cdot C$ and deduce that Γ is co-compact in G . [Hint: Consider the elementary matrices in Γ , and interpret left multiplication by such a matrix as an elementary row operation.]

10 Duality Groups

For any group Γ of type FP , we have seen (6.9) that there is an isomorphism

$$H^n(\Gamma, M) \approx H_0(\Gamma, D \otimes M)$$

for any Γ -module M , where $n = \text{cd } \Gamma$, D is the right Γ -module $H^n(\Gamma, \mathbb{Z}\Gamma)$, and $D \otimes M = D \otimes_{\mathbb{Z}} M$ with the diagonal Γ -action. In this section we will study those groups Γ for which the isomorphism above extends to isomorphisms

$$H^i(\Gamma, M) \approx H_{n-i}(\Gamma, D \otimes M)$$

for all i . These groups were first considered by Bieri and Eckmann [1973], who proved the following characterization of them:

(10.1) Theorem. *The following conditions are equivalent for a group Γ of type FP:*

(i) *There exist an integer n and a Γ -module D such that*

$$H^i(\Gamma, M) \approx H_{n-i}(\Gamma, D \otimes M)$$

for all Γ -modules M and all integers i .

(ii) *There is an integer n such that $H^i(\Gamma, \mathbb{Z}\Gamma \otimes A) = 0$ for all $i \neq n$ and all abelian groups A .*

(iii) *There is an integer n such that $H^i(\Gamma, \mathbb{Z}\Gamma) = 0$ for all $i \neq n$ and $H^n(\Gamma, \mathbb{Z}\Gamma)$ is torsion-free (as an abelian group).*

(iv) *There are natural isomorphisms*

$$H^i(\Gamma, -) \approx H_{n-i}(\Gamma, D \otimes -),$$

where $n = \text{cd } \Gamma$ and $D = H^n(\Gamma, \mathbb{Z}\Gamma)$, which are compatible with the connecting homomorphisms in the long exact homology and cohomology sequences associated to a short exact sequence of modules.

PROOF. (i) $\Rightarrow$ (ii): Apply (i) with M an induced module $\mathbb{Z}\Gamma \otimes A$. It is easy to see that $D \otimes M$ is also induced (cf. III.5.7). Since induced modules are H_* -acyclic, it follows from (i) that $H^i(\Gamma, M) \approx H_{n-i}(\Gamma, D \otimes M) = 0$ for $i \neq n$.

(ii) $\Rightarrow$ (iii): Applying (ii) with $A = \mathbb{Z}$, we find $H^i(\Gamma, \mathbb{Z}\Gamma) = 0$ for $i \neq n$. Now apply (ii) with $A = \mathbb{Z}_k$ ($k > 0$), and use the cohomology exact sequence associated to the short exact sequence

$$0 \rightarrow \mathbb{Z}\Gamma \xrightarrow{k} \mathbb{Z}\Gamma \rightarrow \mathbb{Z}\Gamma \otimes \mathbb{Z}_k \rightarrow 0.$$

This yields

$$0 = H^{n-1}(\Gamma, \mathbb{Z}\Gamma \otimes \mathbb{Z}_k) \rightarrow H^n(\Gamma, \mathbb{Z}\Gamma) \xrightarrow{k} H^n(\Gamma, \mathbb{Z}\Gamma),$$

showing that $H^n(\Gamma, \mathbb{Z}\Gamma)$ has no k -torsion.

(iii) $\Rightarrow$ (iv): Note first that the integer n in (iii) is necessarily equal to $\text{cd } \Gamma$ by 6.7. We now give three different proofs that (iii) $\Rightarrow$ (iv).

First proof: We will use the general nonsense of §III.7, which the reader may want to review before proceeding further. Consider the cohomological functor $H^*(\Gamma, -)$. Since $\text{cd } \Gamma = n < \infty$, we can re-index and regard $H^*(\Gamma, -)$ as a homological functor; indeed, an exact sequence $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ of Γ -modules yields an exact sequence

$$\dots \rightarrow H^{n-1}(\Gamma, M'') \rightarrow H^n(\Gamma, M') \rightarrow H^n(\Gamma, M) \rightarrow H^n(\Gamma, M'') \rightarrow 0,$$

so we obtain a homological functor T by setting $T_i = H^{n-i}(\Gamma, -)$. The hypothesis (iii) implies (via 5.2) that $H^i(\Gamma, F) = 0$ for all $i \neq n$ and all free

$\mathbb{Z}\Gamma$ -modules F . Consequently, T_i is effaceable for $i > 0$. Now consider the functor $H_*(\Gamma, D \otimes -)$, where $D = H^n(\Gamma, \mathbb{Z}\Gamma)$. This is also a homological functor. For if $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ is exact, then so is $0 \rightarrow D \otimes M' \rightarrow D \otimes M \rightarrow D \otimes M'' \rightarrow 0$ (since D is torsion-free by hypothesis), hence we obtain a long exact homology sequence. Moreover, $H_i(\Gamma, D \otimes -)$ is effaceable for $i > 0$ because $D \otimes M$ is induced if M is induced. Since $T_0 \approx H_0(\Gamma, D \otimes -)$ by 6.9, Theorem III.7.3 yields an isomorphism $T \approx H_*(\Gamma, D \otimes -)$ of homological functors, whence (iv).

Second proof: This is similar to the first proof, except that we will use cap products instead of general nonsense to get the map $T \rightarrow H_*(\Gamma, D \otimes -)$. By exercise 5d of §6, there is a “fundamental class” $z \in H_n(\Gamma, D)$ such that cap product with z gives an isomorphism $H^n(\Gamma, -) \xrightarrow{\sim} H_0(\Gamma, D \otimes -)$. Since the cap product map

$$\cap z: T_i = H^{n-i}(\Gamma, -) \rightarrow H_i(\Gamma, D \otimes -)$$

is compatible (up to sign) with connecting homomorphisms in long exact sequences, it follows by dimension-shifting that this map is an isomorphism for all i .

Third proof: Let $P = (P_i)_{0 \leq i \leq n}$ be a finite projective resolution of $\mathbb{Z}$ over $\mathbb{Z}\Gamma$ of length n , and consider the dual complex $\bar{P} = \mathcal{H}om_\Gamma(P, \mathbb{Z}\Gamma)$. Since $H^i(\Gamma, \mathbb{Z}\Gamma) = 0$ for $i \neq n$, $\bar{P}$ provides a projective resolution of $D = H^n(\Gamma, \mathbb{Z}\Gamma)$:

$$\dots \rightarrow \bar{P}^{n-1} \rightarrow \bar{P}^n \rightarrow D \rightarrow 0;$$

more precisely, the n -th suspension $\Sigma^n \bar{P}$ is a projective resolution of D . In view of the duality isomorphism

$$\mathcal{H}om_\Gamma(P, M) \approx \bar{P} \otimes_\Gamma M$$

of I.8.3, it follows that

$$(*) \quad H^i(\Gamma, M) \approx H_{-i}(\bar{P} \otimes_\Gamma M) = H_{n-i}(\Sigma^n \bar{P} \otimes_\Gamma M) = \text{Tor}_{n-i}^\Gamma(D, M)$$

for any Γ -module M . Since D is torsion free, we can now apply III.2.2 to obtain

$$(**) \quad \text{Tor}_{n-i}^\Gamma(D, M) \approx H_{n-i}(\Gamma, D \otimes M).$$

The reader can easily verify that $(*)$ and $(**)$ are natural and compatible with connecting homomorphisms, whence (iv).

(iv) $\Rightarrow$ (i): Trivial. □

If Γ satisfies the conditions of the theorem then Γ is said to be a *duality group* and the Γ -module $D = H^n(\Gamma, \mathbb{Z}\Gamma)$ is called the *dualizing module* of Γ . If, in addition, D is infinite cyclic (as an abelian group), then Γ is said to be a *Poincaré duality group*, and in this case Γ is said to be *orientable* if Γ acts

trivially on D and *non-orientable* otherwise. Note that if Γ is an orientable Poincaré duality group then (iv) takes the familiar form

$$H^i(\Gamma, M) \approx H_{n-i}(\Gamma, M),$$

as in Poincaré duality for closed, orientable manifolds.

Remarks

1. It is sometimes useful to note that if Γ is a duality group, then there are duality isomorphisms $H^i(\Gamma, M) \cong H_{n-i}(\Gamma, D \otimes M)$ given by cap product with a fixed element $z \in H_n(\Gamma, D)$. This was made explicit in the second proof above, and it also follows from the third proof, exactly as in the proof of VI.7.2.

2. Bieri and Eckmann originally defined a duality group to be a group Γ , *not necessarily of type FP*, such that there are cap product isomorphisms

$$\cap z: H^i(\Gamma, M) \cong H_{n-i}(\Gamma, D \otimes M)$$

for all i and M , where n is a fixed integer, D a fixed Γ -module, and z a fixed element of $H_n(\Gamma, D)$. It is clear from the previous remark that a duality group in our sense is also one in theirs. Conversely, if Γ is a duality group in their sense, then naturality of the cap product implies that there are *natural* isomorphisms

$$H^i(\Gamma, -) \approx H_{n-i}(\Gamma, D \otimes -).$$

Since $D \otimes -$ and $H_*(\Gamma, -)$ commute with direct limits, it follows that $H^*(\Gamma, -)$ commutes with direct limits. It also follows that $\text{cd } \Gamma \leq n < \infty$. Thus Γ is of type *FP* by 4.8, and hence Γ is a duality group in our sense. It is also worth noting that the integer n which occurs in the Bieri-Eckmann definition is necessarily equal to $\text{cd } \Gamma$ (this is clear from the proof of 10.1) and that the module D is necessarily isomorphic to $H^n(\Gamma, \mathbb{Z}\Gamma)$. Indeed, the given duality isomorphism with $i = n$ and $M = \mathbb{Z}\Gamma$ gives

$$H^n(\Gamma, \mathbb{Z}\Gamma) \approx H_0(\Gamma, D \otimes \mathbb{Z}\Gamma) \approx D \otimes_{\mathbb{Z}\Gamma} \mathbb{Z}\Gamma \approx D,$$

and this is compatible with the Γ -action by naturality of the duality isomorphism.

EXAMPLES

1. If Γ is a group such that there exists a closed $K(\Gamma, 1)$ -manifold Y , then, as one would expect, Γ is a Poincaré duality group and is orientable if and only if Y is orientable. This follows immediately from 8.1b, 8.2, and the criterion 10.1(iii) for duality. [Alternatively, use Poincaré duality in Y , with local coefficients, to verify directly that Γ satisfies the Bieri-Eckmann definition of duality group which we quoted above, with D equal to the orientation module Ω associated to Y .] For example, the free abelian group $\mathbb{Z}^n$ is an orientable Poincaré duality group. [Take Y to be the n -dimensional torus.] More generally, any finitely generated, torsion-free, nilpotent group

is an orientable Poincaré duality group by example 2 of §9. [Alternatively, one can give a purely algebraic proof, cf. Bieri [1976], Theorem 9.10.] Still more generally, if G is an arbitrary Lie group with only finitely many connected components and Γ is a discrete, co-compact, torsion-free subgroup, then Γ is a Poincaré duality group and is orientable if G is connected (cf. example 4 of §9).

Remark. It is not known whether every Poincaré duality group admits a closed $K(\Gamma, 1)$ -manifold. This is even unknown in the 2-dimensional case. But Eckmann and Müller [1980] have shown that every 2-dimensional Poincaré duality group Γ which has the homology of a surface other than S^2 or P^2 is in fact isomorphic to the fundamental group of that surface. It follows that if there is a 2-dimensional counter-example (i.e., a 2-dimensional Poincaré duality group Γ for which there is no closed $K(\Gamma, 1)$ -manifold, then there is even a counter-example Γ such that $H_*(\Gamma) \approx H_*(S^2)$. Such a Γ would be very interesting indeed. In particular, the method of J. Cohen [1972] shows that Γ would not be of type FL .

2. If Γ is a free group on k generators, $1 \leq k < \infty$, then Γ is a 1-dimensional duality group. Indeed, we know that $\text{cd } \Gamma = 1$ and that Γ is of type FL , so we need only check that $H^0(\Gamma, \mathbb{Z}\Gamma \otimes A) = 0$ for any A ; but this is easily seen to hold for any infinite group Γ , cf. §III.5, exercise 4a. If $k = 1$, then Γ is an orientable Poincaré duality group by example 1 (or by directly computing $H^*(\Gamma, \mathbb{Z}\Gamma)$ from the resolution I.4.5). If $k > 1$, however, then Γ is not a Poincaré duality group; for we have

$$\mathbb{Z}^k \approx H^1(\Gamma, \mathbb{Z}) \approx H_0(\Gamma, D \otimes \mathbb{Z}) = H_0(\Gamma, D),$$

so D cannot be infinite cyclic. (In fact, it is not hard to show that D is a free abelian group of countable rank, cf. exercise 2 below.)

3. The results stated in example 1 of §9 show that knot groups are 2-dimensional duality groups (but not Poincaré duality groups).

4. The results of Borel and Serre quoted in example 5 of §9 show that torsion-free arithmetic groups are duality groups (but not Poincaré duality groups except in the rank 0 case.)

5. If Γ is a finitely presented group of cohomological dimension 2 which cannot be decomposed as a free product, then Γ is a duality group. Indeed, Γ is of type FP by the comments following 6.1, so it suffices to show that $H^i(\Gamma, \mathbb{Z}\Gamma \otimes A) = 0$ for $i < 2$ and all A . This is clear for $i = 0$. For $i = 1$, it is not hard to see (cf. Swan [1969], §3) that $H^1(\Gamma, \mathbb{Z}\Gamma)$ is a free abelian group of rank $e - 1$, where e is the number of “ends” of Γ , and that $H^1(\Gamma, \mathbb{Z}\Gamma \otimes A) \approx H^1(\Gamma, \mathbb{Z}\Gamma) \otimes A$. But $e = 1$ (and hence $H^1(\Gamma, \mathbb{Z}\Gamma \otimes A) = 0$) under our hypotheses, because of the following theorem of Stallings [1968]: A finitely generated, torsion-free group with more than one end is either infinite cyclic or a free product. (See also Scott–Wall [1979] for the theory of ends and a proof of Stallings’s theorem.)

6. Further examples of duality groups can be constructed from known examples by forming extensions and amalgamations. The precise results are somewhat complicated, and we refer to Bieri [1976] for details. [Warning: These group theoretic constructions do *not* always preserve the property of being a duality group. For example, a free product of duality groups is not a duality group unless the groups are free, cf. exercise 3 below.]

Remark. The examples above suggest that if Γ is a duality group then the dualizing module D is either infinite cyclic (in which case Γ is a Poincaré duality group) or free abelian of infinite rank. It is not known whether this is true, the problem being the freeness of D . It is known, however, as a consequence of results of Farrell [1975], that if D is not infinite cyclic then D has infinite rank (i.e., $\dim_{\mathbb{Q}}(D \otimes \mathbb{Q}) = \infty$). A proof can be found in Bieri [1976], §9.5, along with other interesting properties of D .

We close this section by proving one useful property of duality groups:

(10.2) Proposition. *Let Γ be a torsion-free group and Γ' a subgroup of finite index. Then Γ is a duality group if and only if Γ' is a duality group. Moreover, if Γ and Γ' are duality groups then they have the same dualizing module. (More precisely, if D is the dualizing module for Γ , then the dualizing module for Γ' is isomorphic to the module $\text{Res}_{\Gamma'}^{\Gamma} D$ obtained by restricting the action of Γ to Γ' .)*

PROOF. By 6.6, Γ is of type FP if and only if Γ' is of type FP . Since $H^*(\Gamma, \mathbb{Z}\Gamma) \approx H^*(\Gamma', \mathbb{Z}\Gamma')$ by Shapiro's lemma, the first part of the proposition follows from the criterion 10.1(iii) for duality. To prove the second part, we must check that the isomorphism $H^n(\Gamma, \mathbb{Z}\Gamma) \approx H^n(\Gamma', \mathbb{Z}\Gamma')$ is an isomorphism of right Γ' -modules. By straightforward definition-checking (cf. §III.8, exercise 2), one sees that this isomorphism is the composite

$$H^n(\Gamma, \mathbb{Z}\Gamma) \xrightarrow{\text{res}} H^n(\Gamma', \mathbb{Z}\Gamma) \rightarrow H^n(\Gamma', \mathbb{Z}\Gamma'),$$

where the second map is induced by the coefficient homomorphism

$$\mathbb{Z}\Gamma \approx \mathbb{Z}\Gamma \otimes_{\mathbb{Z}\Gamma'} \mathbb{Z}\Gamma' \xrightarrow[(\text{III.5.9})]{\approx} \text{Hom}_{\Gamma'}(\mathbb{Z}\Gamma, \mathbb{Z}\Gamma') \xrightarrow[\text{eval at } 1]{\text{eval at } 1} \mathbb{Z}\Gamma'.$$

One checks that this homomorphism is given explicitly by

$$\gamma \mapsto \begin{cases} \gamma & \gamma \in \Gamma' \\ 0 & \text{otherwise,} \end{cases}$$

and this is clearly compatible with the right Γ' -action. Since $\text{res}: H^n(\Gamma, \mathbb{Z}\Gamma) \rightarrow H^n(\Gamma', \mathbb{Z}\Gamma)$ is obviously compatible with the right Γ' -action (in fact, with the right Γ -action), the proof is complete. $\square$

EXERCISES

1. If Γ is an n -dimensional duality group with dualizing module D , prove that D is a $\mathbb{Z}\Gamma$ -module of type FP (i.e., it admits a finite projective resolution) and of projective dimension n . [Hint: Look at the third proof of (iii) $\Rightarrow$ (iv) in 10.1.]

2. Let Γ be a free group of finite rank > 1 . Prove that the dualizing module $H^1(\Gamma, \mathbb{Z}\Gamma)$ is a free abelian group of infinite rank. [Hint: Use exercise 5 of §5 and argue by induction on the rank of Γ . Alternatively, use the theory of ends.]
3. Show that a duality group of cohomological dimension > 1 cannot be decomposed as a free product. [Hint: Exercise 5 of §5.]
4. If Γ is an n -dimensional duality group whose dualizing module D is $\mathbb{Z}$ -free, prove that

$$H_i(\Gamma, M) \approx H^{n-i}(\Gamma, \text{Hom}(D, M))$$

for any Γ -module M . [Hint: The starting point is exercise 5b of §6. Now imitate any of the three proofs that (iii) $\Rightarrow$ (iv) in 10.1. If you imitate the first or second proof, you will need to use exercise 4b of §5.]

5. Show that if we are willing to use chain complex coefficients (cf. §VII.5) then every group of type FP behaves like a duality group. Namely, there is a “dualizing complex” $\mathcal{D}$ such that

$$H^i(\Gamma, \mathcal{C}) \approx H_{n-i}(\Gamma, \mathcal{C} \otimes \mathcal{D})$$

and

$$H_i(\Gamma, \mathcal{C}) \approx H^{n-i}(\Gamma, \mathcal{H}om(\mathcal{D}, \mathcal{C}))$$

for every complex $\mathcal{C}$ of Γ -modules. [Hint: Let $\mathcal{D}$ be the dual (suitably re-indexed) of a finite projective resolution of $\mathbb{Z}$ over $\mathbb{Z}\Gamma$. Look at the definitions of $H^i(\Gamma, \mathcal{C})$ and $H_i(\Gamma, \mathcal{C})$, and apply exercise 1 of §VI.6.]

- *6. (a) Prove the following result of Strebel [1977]: If Γ is a Poincaré duality group and $\Gamma' \subset \Gamma$ is a subgroup of infinite index, then $\text{cd } \Gamma' < \text{cd } \Gamma$. [Hint: Use exercise 4 of §6.] This result is motivated by the special case (which may in fact be the general case) given in the exercise of §8.
- (b) Give an example which shows that the result of (a) does not hold for arbitrary duality groups.

11 Virtual Notions

Up to now this chapter has been concerned primarily with torsion-free groups. Indeed, we observed in §2 that groups with torsion necessarily have $\text{cd} = \infty$, and hence they certainly cannot satisfy the FL or FP conditions or be duality groups. On the other hand, we have seen examples of groups with torsion (such as $SL_n(\mathbb{Z})$) which have torsion-free subgroups of finite index that satisfy these finiteness conditions. The purpose of this section is to introduce a convenient language for describing this situation.

In general we will say that a group *virtually* has a given property if some subgroup of finite index has that property. For instance, Γ is *virtually torsion-free* if Γ has a torsion-free subgroup of finite index. In this case it follows from

Serre's theorem (3.1) that all such subgroups have the same cohomological dimension (which may be finite or infinite); for if Γ' and Γ'' are two torsion-free subgroups of finite index, then $\Gamma' \cap \Gamma''$ has finite index in both Γ' and Γ'' , so Serre's theorem gives $\text{cd } \Gamma' = \text{cd}(\Gamma' \cap \Gamma'') = \text{cd } \Gamma''$. The common cohomological dimension of the torsion-free subgroups of finite index is called the *virtual cohomological dimension* of Γ and is denoted $\text{vcd } \Gamma$.

Similarly, Γ is said to be of type *VFP* (resp. *VFL*) if some subgroup of finite index is of type *FP* (resp. *FL*). If Γ is of type *VFP* then it follows from 6.6 that every torsion-free subgroup of finite index is of type *FP*. The corresponding statement for groups of type *VFL* is not known, so one is led to introduce the following apparently stronger condition: Γ is said to be of type *WFL* if Γ is virtually torsion-free and every torsion-free subgroup of finite index is of type *FL*.

To summarize, we have introduced four "virtual" finiteness conditions in this section, which are related as follows:

$$WFL \Rightarrow VFL \Rightarrow VFP \Rightarrow (\text{vcd} < \infty).$$

It is not known whether the first two implications are reversible. Note also that $VFP \Rightarrow FP_\infty$ by 5.1.

We saw in §7 that $\text{cd } \Gamma < \infty$ if and only if there exists a finite dimensional, contractible, free Γ -CW-complex. To state the "virtual" analogue of this, we introduce the following weakening of the notion of "free action": We say that a Γ -complex is *proper* if the isotropy group Γ_σ is finite for every cell σ . (This is consistent with the notion of "proper action" introduced in §II.4, Example 6, but we will not need to know this.) Note that a proper Γ -complex is Γ' -free for every torsion-free subgroup $\Gamma' \subseteq \Gamma$. Conversely, if X is a Γ -complex which is Γ' -free for some subgroup Γ' of finite index (i.e., if X is *virtually free*), then X is proper. Thus, if Γ is virtually torsion-free, a Γ -complex is proper if and only if it is virtually free.

We can now state the topological interpretation of finite virtual cohomological dimension:

(11.1) Theorem. *Let Γ be a virtually torsion-free group. Then $\text{vcd } \Gamma < \infty$ if and only if there exists a finite dimensional, contractible, proper Γ -complex X . If there exists such an X which has only finitely many cells mod Γ , then Γ is of type *WFL*.*

PROOF. The second assertion and the "if" part of the first assertion are obvious from the remarks above. The "only if" part follows from the proof of Serre's theorem (3.1). Indeed, the entire proof of that theorem, except for the last sentence, is valid for any group Γ with $\text{vcd } \Gamma < \infty$, and it yields the desired X . $\square$

There remain several open questions in connection with this theorem. It is not known, for example, whether X can be taken to have dimension equal to

$\text{vcd } \Gamma$. Clearly we have $\text{vcd } \Gamma \leq \dim X$ for any X as in 11.1; but the proof above *always* yields an X with $\dim X > \text{vcd } \Gamma$, except in the trivial case where Γ is finite and X is a point. It is also not known whether the converse of the second assertion of 11.1 is true. In particular, the proof above will always yield an X with infinitely many cells mod Γ , unless Γ is finite. Thus we are very far from understanding the relation between our algebraically defined finiteness conditions and the analogous topologically defined conditions.

For future reference, we record now some facts which can be deduced from the proof of 11.1:

(11.2) Addendum. *If $\text{vcd } \Gamma < \infty$ then the complex X in 11.1 can be taken to be simplicial (with simplicial Γ -action) and to have the following property: For every finite subgroup $H \subset \Gamma$, the fixed-point set X^H is non-empty and contractible.*

PROOF. The assertion about the fixed-point sets follows easily from the proof of 11.1, cf. §3, exercise. The fact that X can be taken to be simplicial requires some preliminaries concerning ordered simplicial complexes.

An *ordered simplicial complex* is a simplicial complex together with a partial ordering on its vertices, such that the vertices of any simplex are linearly ordered: $v_0 < v_1 < \dots < v_n$. The canonical example of an ordered simplicial complex is the barycentric subdivision K' of an arbitrary simplicial complex K . To see that K' is ordered, recall that the vertices of K' are the barycenters of the simplices of K , hence there is an ordering on the vertices of K' corresponding to the obvious ordering (by the face relation) on the simplices of K ; moreover, the simplices of K' are precisely the finite sets of barycenters which are linearly ordered (cf. Spanier [1966], 3.3, or Schubert [1968], III.2.6), so K' is indeed an ordered simplicial complex.

Next we recall the definition of the *simplicial product* $K \times L$ of two ordered simplicial complexes K and L . The set of vertices $\text{vert}(K \times L)$ is defined by $\text{vert}(K \times L) = \text{vert}(K) \times \text{vert}(L)$, with the product ordering: $(v, w) \leq (v', w') \Leftrightarrow v \leq v' \text{ and } w \leq w'$. A simplex of $K \times L$ is defined to be a linearly ordered set of vertices $(v_0, w_0) < \dots < (v_n, w_n)$ such that $\{v_0, \dots, v_n\}$ is a simplex of K (possibly of dimension $< n$) and $\{w_0, \dots, w_n\}$ is a simplex of L . It is a classical fact (cf. Eilenberg–Steenrod [1952], II.8.9, or Milnor [1957b]) that this abstract product complex $K \times L$ provides a triangulation of the geometric product of K and L , i.e., that $|K \times L| \approx |K| \times |L|$, where $| \cdot |$ denotes geometric realization. [Here, as in the proof of 3.1, some care needs to be taken in topologizing $|K| \times |L|$.]

Look now at the proof of 3.1. The $K(\Gamma', 1)$ -complex Y' can be taken to be simplicial by 7.3. Its universal cover X' is then simplicial (with simplicial Γ' -action), cf. Spanier [1966], Theorem 3.8.3, or Schubert [1968], §III.6.9. Passing to the barycentric subdivision if necessary, we can assume that X' is an *ordered* simplicial complex and that the Γ' -action preserves the ordering. The co-induction construction given in the proof of 3.1 can therefore be done in the

category of ordered simplicial complexes using the product described above, and the resulting X will then be simplicial. $\square$

EXAMPLES

1. $\text{vcd } \Gamma = 0$ if and only if Γ is finite.

2. If $\Gamma = \Gamma_1 *_A \Gamma_2$ where Γ_1 and Γ_2 are finite, then Γ is of type *WFL* and $\text{vcd } \Gamma \leq 1$. More generally, the same results are true whenever Γ is the fundamental group of a finite graph of finite groups, cf. §VII.9. This is an easy consequence of Serre's theory of groups acting on trees, which provides a contractible 1-complex on which Γ acts properly and with finite quotient. For details see Serre [1977a], II.2.6, Prop. 11. [Note: If we drop the requirement that the graph of finite groups be finite and require instead that there be a bound on the orders of the finite groups, then it is still true that $\text{vcd } \Gamma \leq 1$, but Γ will not necessarily be of type *WFL*.] Conversely, if Γ is a group such that $\text{vcd } \Gamma \leq 1$, then the Stallings–Swan theorem (§2, Example 2) shows that Γ has a free subgroup of finite index, and it is then known that Γ is the fundamental group of a graph of finite groups. This is due to Karrass–Pietrowski–Solitar [1973] in case Γ is finitely generated (in which case the graph can be taken to be finite) and to D. E. Cohen [1972] and Scott [1974] in the general case. Proofs can also be found in Dunwoody [1979] and Scott–Wall [1979].

3. If Γ is a finitely generated one-relator group, then Γ is of type *WFL* and $\text{vcd } \Gamma \leq 2$. Indeed, Γ is virtually torsion-free by Fischer–Karrass–Solitar [1972]. And Lyndon [1950] established an exact sequence

$$0 \rightarrow \mathbb{Z}[\Gamma/C] \rightarrow F \rightarrow \mathbb{Z}\Gamma \rightarrow \mathbb{Z} \rightarrow 0,$$

where F is a free $\mathbb{Z}\Gamma$ -module of finite rank and C is a finite cyclic subgroup of Γ . Our assertions follow at once, since $\mathbb{Z}[\Gamma/C]$ is clearly a free $\mathbb{Z}\Gamma'$ -module of finite rank for every torsion-free subgroup $\Gamma' \subseteq \Gamma$ of finite index. [Note that we have in this case a contractible, proper, 2-dimensional Γ -complex with finite quotient. This follows from the topological version of Lyndon's theorem, which we stated in Exercise 2c of §II.5.]

4. Let G be a Lie group and let X be the associated homogeneous space G/K , where K is a maximal compact subgroup. Let Γ be a discrete subgroup of G , and assume that Γ is virtually torsion-free. [This is automatic if, for instance, Γ is a subgroup of some $GL_n(\mathbb{Z})$, cf. §II.4, Exercise 3.] Then it follows from Example 4 of §9 that

$$\text{vcd } \Gamma \leq \dim X,$$

with equality if and only if Γ is co-compact in G . In the latter case, Γ is of type *WFL*. More interestingly, Γ is of type *WFL* in the arithmetic case discussed in

Example 5 of §9, even though Γ is usually not co-compact, and the Borel–Serre results which we stated there give a precise formula for $\text{vcd } \Gamma$. For instance

$$\text{vcd } SL_n(\mathbb{Z}) = \frac{n(n-1)}{2}.$$

Finally, a group Γ is said to be a *virtual duality group* if some subgroup of finite index is a duality group. The arithmetic groups provide a large and interesting class of virtual duality groups.

(11.3) Proposition. *A group Γ is a virtual duality group if and only if the following two conditions are satisfied:*

- (a) Γ is of type *VFP*.
- (b) *There is an integer n such that $H^i(\Gamma, \mathbb{Z}\Gamma) = 0$ for $i \neq n$ and $H^n(\Gamma, \mathbb{Z}\Gamma)$ is torsion-free.*

In this case every torsion-free subgroup of finite index is a duality group with dualizing module $H^n(\Gamma, \mathbb{Z}\Gamma)$.

This is an easy consequence of Shapiro’s lemma, as in the proof of 10.2.

EXERCISES

1. Suppose Γ is virtually torsion-free and let Γ' be an arbitrary subgroup. Show that $\text{vcd } \Gamma' \leq \text{vcd } \Gamma$, with equality if $(\Gamma : \Gamma') < \infty$.
- *2. Study the behavior of the virtual finiteness properties with respect to extensions, amalgamations, etc. [Hint: Choose a torsion-free subgroup of finite index and apply 2.4b, Exercise 4 of §2, and Exercise 8 of §6.] *Warning:* These group-theoretic constructions do *not* preserve the property of being virtually torsion-free, cf. Schneebeli [1978]. So the results you state will have to include a hypothesis that the group under consideration is virtually torsion-free.

CHAPTER IX

Euler Characteristics

1 Ranks of Projective Modules: Introduction

Let Γ be a finite group and P a finitely generated projective $\mathbb{Z}\Gamma$ -module. There are several ways that one might try to associate a “rank” to P . One candidate for this, which we will denote by $\varepsilon(P)$ or $\varepsilon_\Gamma(P)$, is defined via extension of scalars with respect to the augmentation map $\mathbb{Z}\Gamma \rightarrow \mathbb{Z}$. Namely, we consider $P_\Gamma = \mathbb{Z} \otimes_{\mathbb{Z}\Gamma} P$, which is a finitely generated projective $\mathbb{Z}$ -module; since projective $\mathbb{Z}$ -modules are free, we can set

$$(1.1) \quad \varepsilon(P) = \text{rk}_{\mathbb{Z}}(P_\Gamma),$$

the right hand side being the rank in the naive sense (i.e., the cardinality of a basis). Note that 1.1 makes sense even if Γ is infinite.

Our second definition of a “rank” of P , which we will denote $\rho(P)$ or $\rho_\Gamma(P)$, is based on the fact that P itself is finitely generated and projective as a $\mathbb{Z}$ -module; hence we can set

$$(1.2) \quad \rho(P) = \text{rk}_{\mathbb{Z}}(P)/|\Gamma|.$$

Here, of course, we do need Γ to be finite.

Both ε and ρ give the “right” answer n if P is the free module $\mathbb{Z}\Gamma^n$. Other than that, however, it is not obvious that they have anything to do with one another. In fact, it is not even obvious that the rational number $\rho(P)$ is an integer. But it turns out that $\varepsilon = \rho$ for all finite groups Γ and all finitely generated projectives P . This fact, which is due to Swan, plays a crucial role in our theory of Euler characteristics. We will therefore devote the first few sections of the chapter to a proof of this equality.

Following Bass [1976, 1979], our proof will go as follows: In §2 we will introduce a third notion of rank, due to Hattori and Stallings and defined for

finitely generated projectives over an arbitrary ring. In §3 we will use standard commutative algebra to give a concrete interpretation of the Hattori–Stallings rank in case the ring is commutative. This interpretation, together with formal properties of the Hattori–Stallings rank over group rings, will lead in §4 to a proof that $\varepsilon = \rho$.

EXERCISE

The purpose of this exercise is to show that some restriction on a ring A is needed if there is to be a “reasonable” $\mathbb{Z}$ -valued rank for finitely generated projective A -modules. Suppose there is a $\mathbb{Z}$ -valued function r on finitely generated projective A -modules, satisfying:

- (i) $r(P \oplus Q) = r(P) + r(Q)$.
- (ii) $r(P) > 0$ if $P \neq 0$.
- (iii) $r(A) = 1$.

Prove then that A is *indecomposable*, i.e., that A cannot be decomposed as the direct sum of two non-zero left ideals. [Hint: A decomposition $A = I \oplus J$ yields an equation $1 = r(I) + r(J)$ in $\mathbb{Z}$.]

2 The Hattori–Stallings Rank

If F is a finitely generated free $\mathbb{Z}$ -module, then the rank of F is equal to the trace of the identity endomorphism of F ; indeed, this is just a restatement of the obvious fact that the trace of the $n \times n$ identity matrix is n . With this as motivation, Hattori [1965] and Stallings [1965b] defined the “rank” of an arbitrary finitely generated projective module over an arbitrary ring to be the trace of the identity map. In order to make sense out of this, we need to begin by developing a theory of traces for endomorphisms of projectives which are not necessarily free over rings which are not necessarily commutative.

Let A be an arbitrary ring, F a finitely generated free A -module, and $\alpha: F \rightarrow F$ an endomorphism. One would like to define $\text{tr}(\alpha)$ as the sum of the diagonal entries of the matrix of α relative to a basis for F . Unfortunately, this sum is not in general independent of the choice of basis for F , unless A is commutative. (One can already see this for 1×1 matrices.)

There is, however, a well-defined trace in the quotient $T(A) = A/[A, A]$, where $[A, A]$ is the additive subgroup of A generated by all commutators $ab - ba$, $a, b \in A$. [Warning: $[A, A]$ is not an ideal, in general, so $T(A)$ is simply an abelian group, not a ring.] Namely, we define $\text{tr}(\alpha) \in T(A)$ for α an

$n \times n$ matrix to be $\sum_{i=1}^n \bar{\alpha}_{ii}$, where $\bar{\alpha}_{ii}$ is the image of α_{ii} in $T(A)$. It is trivial to verify that

$$(2.1) \quad \text{tr}(\alpha\beta) = \text{tr}(\beta\alpha)$$

whenever α is an $m \times n$ matrix and β is an $n \times m$ matrix (so that $\alpha\beta$ and $\beta\alpha$ are both square); and from this it follows that $\text{tr}(\beta\alpha\beta^{-1}) = \text{tr}(\beta^{-1}\beta\alpha) = \text{tr}(\alpha)$ for any invertible β , so the trace of an endomorphism is indeed a well-defined element of $T(A)$, independent of the choice of basis.

We would like now to extend the trace to endomorphisms of finitely generated *projective* modules. I claim that there is a unique way to do this so that 2.1 continues to hold for maps $P \xrightarrow{\alpha} P'$. For let $\alpha: P \rightarrow P$ be an endomorphism of a finitely generated projective, and express P as a direct summand of a finitely generated free module F via maps $P \xrightarrow{\pi} F$ with $\pi_1 = \text{id}_P$. Then α gives rise to an endomorphism $\iota\alpha\pi$ of F . [In concrete terms, this is simply α on the summand P and 0 on the complementary summand $\ker \pi$.] If we want 2.1 to hold, then we are forced to define

$$(2.2) \quad \text{tr}(\alpha) = \text{tr}(\iota\alpha\pi);$$

for 2.1 would imply $\text{tr}(\iota\alpha\pi) = \text{tr}(\alpha\pi_1) = \text{tr}(\alpha \circ \text{id})$.

It remains to show that the right side of 2.2 is independent of the choice of $P \hookrightarrow F$ and that 2.1 continues to hold. Suppose we have maps

$$F \xrightarrow{\pi} P \xrightarrow{\alpha} P' \xrightarrow{\pi'} F',$$

where F and F' are free, $\pi_1 = \text{id}_P$, and $\pi'_1 = \text{id}_{P'}$. Then we can apply 2.1 to matrices representing the composites

$$F \xrightleftharpoons[\iota'\beta\pi]{\iota\alpha\pi'} F'$$

to conclude that $\text{tr}(\iota\alpha\pi'\iota'\beta\pi) = \text{tr}(\iota'\beta\pi\iota\alpha\pi')$. Since π'_1 and π_1 are identity maps, this yields

$$\text{tr}(\iota\alpha\beta\pi) = \text{tr}(\iota'\beta\alpha\pi').$$

Taking $P' = P$ and $\beta = \text{id}$, we conclude that the right side of 2.2 is indeed independent of the choice of $P \hookrightarrow F$. And taking P' and β arbitrary, we conclude that 2.1 holds for arbitrary $P \xrightarrow{\alpha} P'$. This proves the claim. The resulting trace $\text{tr}(\alpha)$ will sometimes be denoted $\text{tr}_A(\alpha)$.

Next we discuss the behavior of traces with respect to *extension and restriction of scalars*.

(2.3) Proposition. *Let $\varphi: A \rightarrow B$ be a ring homomorphism and let $T(\varphi): T(A) \rightarrow T(B)$ be the induced map. If $\alpha: P \rightarrow P$ is an endomorphism of a finitely generated projective A -module, then $B \otimes_A \alpha: B \otimes_A P \rightarrow B \otimes_A P$ is an endomorphism of a finitely generated projective B -module, and we have*

$$\text{tr}_B(B \otimes_A \alpha) = T(\varphi)(\text{tr}_A(\alpha)).$$

PROOF. It suffices to consider the case where P is free. Then $B \otimes_A P$ is also free, and a matrix representing $B \otimes_A \alpha$ is obtained by applying φ to the entries of a matrix representing α . $\square$

(2.4) Proposition. *Let $\varphi: A \rightarrow B$ be a ring homomorphism such that B is finitely generated and projective as a left A -module. Then there is a map*

$$\mathrm{tr}_{B/A}: T(B) \rightarrow T(A)$$

with the following property: Let $\alpha: P \rightarrow P$ be an endomorphism of a finitely generated projective B -module; then P is also finitely generated and projective as an A -module, and

$$\mathrm{tr}_A(\alpha) = \mathrm{tr}_{B/A}(\mathrm{tr}_B(\alpha)).$$

The proof will use:

(2.5) Lemma. *Let A be an arbitrary ring and let P be a finitely generated projective A -module which is a direct sum $P_1 \oplus \cdots \oplus P_n$. Let $\alpha: P \rightarrow P$ be an endomorphism with components $\alpha_{ij}: P_i \rightarrow P_j$. Then $\mathrm{tr}(\alpha) = \sum_1^n \mathrm{tr}(\alpha_{ii})$.*

PROOF. This reduces easily to the case where each P_i is free, in which case the result follows from an examination of the matrix of α . $\square$

PROOF OF 2.4. For any $b \in B$, the right multiplication map $\mu_b: B \rightarrow B$ is an endomorphism of the left A -module B , so it has a trace $\mathrm{tr}_A(\mu_b) \in T(A)$. Consider the additive map $B \rightarrow T(A)$ given by $b \mapsto \mathrm{tr}_A(\mu_b)$. Since $\mu_{b_1 b_2} = \mu_{b_2} \mu_{b_1}$, this map annihilates commutators $b_1 b_2 - b_2 b_1$. There is therefore an induced map $T(B) \rightarrow T(A)$, which is the desired $\mathrm{tr}_{B/A}$. To prove the formula $\mathrm{tr}_A(\alpha) = \mathrm{tr}_{B/A}(\mathrm{tr}_B(\alpha))$, note first that it holds if $P = B$ (in which case α is necessarily of the form μ_b). In view of Lemma 2.5, it continues to hold if P is a direct sum of copies of B , i.e., if P is a free B -module. In the general case, choose $P \xrightarrow{\pi} F$ with $\pi i = \mathrm{id}_P$, where F is free over B . Then $\mathrm{tr}(\alpha \pi) = \mathrm{tr}(\alpha \pi i) = \mathrm{tr}(\alpha)$ over both A and B , so the formula for $\alpha: P \rightarrow P$ follows from the formula for $\alpha \pi: F \rightarrow F$. $\square$

We can now define the *Hattori–Stallings rank* of P , denoted $R(P)$ or $R_A(P)$, by

$$R(P) = \mathrm{tr}(\mathrm{id}_P).$$

Note that $R(P)$ is not a number, but rather an element of the somewhat mysterious abelian group $T(A)$. In the next two sections, however, we will give concrete interpretations of $R(P)$ for some interesting classes of rings. Before proceeding to this, the reader is strongly urged to look at the exercises below; the results of exercises 5, 6, and 7 will be needed in §4.

EXERCISES

1. Let P be a finitely generated projective (left) A -module and let P^* be its dual. Then P^* is a right A -module and we have $P^* \otimes_A P \approx \text{Hom}_A(P, P)$ by 1.8.3b. Prove that there is a commutative diagram

$$\begin{array}{ccc} P^* \otimes_A P & \approx & \text{Hom}_A(P, P) \\ \text{ev} \searrow & & \swarrow \text{tr} \\ & T(A) & \end{array}$$

where ev , the “evaluation map,” is given by $\text{ev}(u \otimes x) = \overline{u(x)}$. In some treatments of the theory of traces, this diagram is used to define tr , cf. Bass [1976, 1979]. [Hint: Consider $P \xrightarrow{\sim} \frac{A}{I} A$.]

2. (a) Let C be a finite projective chain complex over A and let $\tau: C \rightarrow C$ be a chain map with components $\tau_i: C_i \rightarrow C_i$. Define the “Lefschetz number” $L(\tau) \in T(A)$ by $L(\tau) = \sum (-1)^i \text{tr}(\tau_i)$. Prove that $L(\tau_0) = L(\tau_1)$ if $\tau_0 \simeq \tau_1$. [Hint: Write down the definition of homotopy and use 2.1.]
- (b) Deduce that one can define a trace map $\text{tr}: \text{Hom}_A(M, M) \rightarrow T(A)$ whenever M is an A -module of type FP by setting $\text{tr}(\alpha) = L(\tau)$, where τ is any lifting of α to a finite projective resolution of M .
3. Suppose A is an algebra over a commutative ring k (e.g., A is always an algebra over its center). Show that $\text{Hom}_A(P, P)$ and $T(A)$ are k -modules and that $\text{tr}: \text{Hom}_A(P, P) \rightarrow T(A)$ is k -linear. More generally, show that the same is true if P is replaced by a module M of type FP as in exercise 2b.
4. Show by checking definitions that $R(P)$ is equal to the trace of an idempotent matrix defining P . More precisely, let F be a finitely generated free module and $e: F \rightarrow F$ an (idempotent) projection operator of F onto a direct summand isomorphic to P ; then $R(P) = \text{tr}(e)$.
5. Let Γ be a group and $\varphi: \mathbb{Z}\Gamma \rightarrow \mathbb{Z}$ the augmentation map. Show that the numerical “rank” $e_\Gamma(\quad)$ defined in §1 can be obtained from the Hattori–Stallings rank $R_{\mathbb{Z}\Gamma}(\quad)$ by the formula

$$e_\Gamma(P) = T(\varphi)(R_{\mathbb{Z}\Gamma}(P)).$$

[Apply 2.3 with $\alpha = \text{id}_P$ and note that $T(\varphi)$ takes values in $\mathbb{Z}$ because $T(\mathbb{Z}) = \mathbb{Z}$.]

6. Let Γ be a finite group.

- (a) Look at the definition of $\text{tr}_{\mathbb{Z}\Gamma/\mathbb{Z}}: T(\mathbb{Z}\Gamma) \rightarrow T(\mathbb{Z}) = \mathbb{Z}$ in the proof of 2.4 and deduce that $\text{tr}_{\mathbb{Z}\Gamma/\mathbb{Z}}(\bar{1}) = |\Gamma|$ and that $\text{tr}_{\mathbb{Z}\Gamma/\mathbb{Z}}(\bar{\gamma}) = 0$ for $1 \neq \gamma \in \Gamma$. Here $\bar{\gamma}$ is the image of γ in $T(\mathbb{Z}\Gamma)$. Consequently, there is a well-defined homomorphism $\tau: T(\mathbb{Z}\Gamma) \rightarrow \mathbb{Z}$ such that $\tau(\bar{1}) = 1$ and $\tau(\bar{\gamma}) = 0$ for $1 \neq \gamma \in \Gamma$, and one has $\text{tr}_{\mathbb{Z}\Gamma/\mathbb{Z}} = |\Gamma| \cdot \tau$.
- (b) Deduce from 2.4 applied to the inclusion $\mathbb{Z} \hookrightarrow \mathbb{Z}\Gamma$ (with $\alpha = \text{id}_P$) that $\text{rk}_{\mathbb{Z}}(P) = |\Gamma| \cdot \tau(R_{\mathbb{Z}\Gamma}(P))$ for any finitely generated projective $\mathbb{Z}\Gamma$ -module P . Hence the “rank” ρ defined in §1 is given by

$$\rho_\Gamma(P) = \tau(R_{\mathbb{Z}\Gamma}(P)).$$

7. Exercise 6b shows that $\rho_\Gamma(P) \in \mathbb{Z}$. On the other hand, it is obvious from the definition of ρ that $\rho_\Gamma(P) > 0$ if $P \neq 0$. Show, therefore, that $\mathbb{Z}\Gamma$ is indecomposable when Γ is finite. [Use the exercise of §1.]
8. Show that the definition of τ in exercise 6a makes sense even if Γ is infinite, i.e., there is a well-defined homomorphism $\tau: T(\mathbb{Z}\Gamma) \rightarrow \mathbb{Z}$ such that $\tau(\bar{1}) = 1$ and $\tau(\bar{\gamma}) = 0$ for $1 \neq \gamma \in \Gamma$. We can therefore define a $\mathbb{Z}$ -valued “rank” ρ over $\mathbb{Z}\Gamma$ by $\rho(P) = \tau(R(P))$. In view of exercise 6b, this agrees with the ρ of §1 if Γ is finite. Moreover, it turns out for arbitrary Γ that $\rho(P) > 0$ if $P \neq 0$. [This non-trivial result is due to Kaplansky; proofs can be found in Montgomery [1969] and in Passman [1977], Theorem 1.8 and Exercise 9 of Chapter 1.] Consequently, you can show as in exercise 7 that $\mathbb{Z}\Gamma$ is indecomposable for *any* group Γ .

3 Ranks over Commutative Rings

If A is an integral domain with field of fractions K , then there is an easy way to define a $\mathbb{Z}$ -valued rank for finitely generated projective A -modules. Namely, we set

$$\text{rk}_A(P) = \dim_K(K \otimes_A P).$$

Moreover, the Hattori–Stallings rank $R_A(P)$ is simply given by

$$(3.1) \quad R_A(P) = \text{rk}_A(P) \cdot 1,$$

where $1 \in A = T(A)$ is the identity element. [To prove this, we may regard both sides of the equation as elements of K . The left hand side is then equal to $R_K(K \otimes_A P)$ by the compatibility of traces with extension of scalars (2.3), and this in turn is equal to $\dim_K(K \otimes_A P) \cdot 1 = \text{rk}_A(P) \cdot 1$ since $K \otimes_A P$ is a free K -module.]

Our goal in this section is to prove a generalization of 3.1 to indecomposable commutative rings which are not necessarily domains. [This restriction to indecomposable rings should not surprise the reader in view of the exercise of §1. There are, however, interesting results along these lines for decomposable rings; see the exercises below.]

Since A is no longer assumed to be a domain, we do not have a field of fractions to work with. We do, however, have a localization $A_{\mathfrak{p}}$ for every prime ideal $\mathfrak{p}$, and we can use this instead. [See Atiyah–Macdonald [1969], Chapter 3, for the definitions and elementary facts concerning localization.] The ring $A_{\mathfrak{p}}$ is a *local ring* (it has a unique maximal ideal), and we will prove:

(3.2) Proposition. *Any finitely generated projective over a local ring is free.*

We can therefore define the *rank of P at $\mathfrak{p}$* to be the rank in the naive sense of the free $A_{\mathfrak{p}}$ -module $P_{\mathfrak{p}} = A_{\mathfrak{p}} \otimes_A P$. Let $\text{Spec } A$ be the set of prime ideals

of A . Then what we have, so far, is a function $\mathcal{R}_A(P): \text{Spec } A \rightarrow \mathbb{Z}$ whose value at $\mathfrak{p}$ is the rank of P at $\mathfrak{p}$.

$\text{Spec } A$ has a well-known topology (the “Zariski topology”), whose definition we will recall below, and we will prove:

(3.3) Proposition. *The function $\mathcal{R}_A(P)$ is locally constant.*

Finally, we will establish the (not surprising) geometric interpretation of indecomposability:

(3.4) Proposition. *If A is indecomposable then $\text{Spec } A$ is connected.*

The function $\mathcal{R}_A(P)$ is therefore constant if A is indecomposable, and we set $\text{rk}_A(P)$ equal to its constant value.

Accepting all this for the moment, we can prove the desired generalization of 3.1:

(3.5) Theorem. *Let A be a commutative indecomposable ring, let P be a finitely generated projective A -module, and let $\text{rk}_A(P)$ be the rank of P at any prime ideal $\mathfrak{p}$. Then the Hattori–Stallings rank $R_A(P)$ is given by $R_A(P) = \text{rk}_A(P) \cdot 1$.*

PROOF. The compatibility of traces with extension of scalars implies, as in the proof of 3.1, that $R_A(P)$ and $\text{rk}_A(P) \cdot 1$ have the same image in $A_{\mathfrak{p}}$ for every $\mathfrak{p}$; hence they are equal. $\square$

It remains to prove 3.2, 3.3, and 3.4. For 3.2 we will need “Nakayama’s lemma”:

(3.6) Lemma. *Let A be a local ring with maximal ideal $\mathfrak{m}$ and residue field $k = A/\mathfrak{m}$, and let M be a finitely generated A -module. If $k \otimes_A M = 0$ then $M = 0$.*

(Note the similarity between this and VI.8.4. Both results are special cases of a general “Nakayama’s lemma.”)

PROOF. Let $m_1, \dots, m_r$ be a minimal set of generators of M , and suppose $r \geq 1$. Since $k \otimes_A M = M/\mathfrak{m}M$, our hypothesis says that $M = \mathfrak{m}M$. In particular, we can write $m_1 = \sum_1^r a_i m_i$ with $a_i \in \mathfrak{m}$, so $(1 - a_1)m_1 = \sum_2^r a_i m_i$. But $1 - a_1$ is invertible since it is not in the unique maximal ideal of A ; thus M is generated by $m_2, \dots, m_r$. This contradicts the minimality of $m_1, \dots, m_r$, so r must be 0. $\square$

PROOF OF 3.2. This is similar to the proof of VI.8.5. Let A be local with residue field k , let P be a finitely generated projective A -module, and let

$$r = \dim_k(k \otimes_A P).$$

Then we can find a map $f: A' \rightarrow P$ such that

$$k \otimes f: k' = k \otimes A' \rightarrow k \otimes P$$

is an isomorphism. Then $k \otimes \text{coker } f = \text{coker}(k \otimes f) = 0$, so $\text{coker } f = 0$ by 3.6. We now have a short exact sequence

$$0 \rightarrow \ker f \rightarrow A' \xrightarrow{f} P \rightarrow 0,$$

which splits because P is projective. The sequence therefore remains exact after being tensored with k , so $k \otimes \ker f = 0$. Now $\ker f$ is finitely generated, being a direct summand of A' , so we can apply 3.6 again to conclude that $\ker f = 0$. Thus f is an isomorphism. $\square$

The next step is to topologize $\text{Spec } A$. For any subset $S \subseteq A$, we define a subset $V(S) \subseteq \text{Spec } A$ by

$$V(S) = \{p: p \supseteq S\}.$$

(3.7) Lemma.

- (i) If I is the ideal generated by S , then $V(S) = V(I)$.
- (ii) $V(S) = \text{Spec } A$ if and only if every element of S is nilpotent.
- (iii) If I is an ideal, then $V(I) = \emptyset$ if and only if $I = A$.
- (iv) If I and J are ideals, then $V(IJ) = V(I) \cup V(J)$.
- (v) If $\{I_\alpha\}$ is a family of ideals and I is their sum, then $V(I) = \bigcap_\alpha V(I_\alpha)$.

PROOF. The proofs are routine verifications which are left for the reader, except for the “only if” part of (ii), which is proved as follows: Suppose $a \in S$ and a is not nilpotent. Then the localization $A[a^{-1}]$ is not the zero ring and hence contains a maximal ideal $\mathfrak{m}$. The contraction of $\mathfrak{m}$ to A is then a prime ideal not containing a , so $V(S) \neq \text{Spec } A$. $\square$

It follows from (ii)–(v) that we can topologize $\text{Spec } A$ by declaring that the closed sets are the sets of the form $V(I)$, where I is an ideal of A . In view of (i), every $V(S)$ is then closed.

PROOF OF 3.3. Suppose P_p is free of rank r over A_p . We must show that P_q is free of rank r over A_q for all q in some neighborhood of p . It is easy to see that we can find r elements of P whose images in P_p form a basis for P_p . In other words, we can find a map $f: F \rightarrow P$, where F is a free A -module of rank r , such that $f_p: F_p \rightarrow P_p$ is an isomorphism. Therefore $(\text{coker } f)_p = 0$, so every element of $\text{coker } f$ is annihilated by an element of $A - p$. But $\text{coker } f$ is finitely generated, so there is a single element $s \in A - p$ such that $s \cdot \text{coker } f = 0$. Thus if we localize A by inverting s , we obtain a surjection $f[s^{-1}]: F[s^{-1}] \rightarrow P[s^{-1}]$ which becomes an isomorphism when localized at p . (This makes sense since A_p can be viewed as a localization of $A[s^{-1}]$.)

$P[s^{-1}]$ being projective over $A[s^{-1}]$, this surjection splits, and $\ker f[s^{-1}]$ is therefore finitely generated. Since it becomes zero when localized at p ,

it is annihilated by some element of $A[s^{-1}] - \mathfrak{p}[s^{-1}]$ and hence by some $t \in A - \mathfrak{p}$. Letting $u = st$, we conclude that $F[u^{-1}] \xrightarrow{\sim} P[u^{-1}]$. But then $F_{\mathfrak{q}} \xrightarrow{\sim} P_{\mathfrak{q}}$ for all $\mathfrak{q}$ such that $u \notin \mathfrak{q}$, since $A_{\mathfrak{q}}$ is a localization of $A[u^{-1}]$ for such $\mathfrak{q}$. Thus $P_{\mathfrak{q}}$ is free of rank r for all $\mathfrak{q}$ in the neighborhood $\text{Spec } A - V(u)$ of $\mathfrak{p}$. $\square$

PROOF OF 3.4. Note first that a ring A is decomposable if and only if it contains an idempotent element $e \neq 0, 1$; for direct sum decompositions of the A -module A correspond to 1×1 idempotent matrices, which are the same as idempotent elements. What we must prove, then, is that if $\text{Spec } A$ is disconnected then A contains an idempotent $e \neq 0, 1$.

Suppose $\text{Spec } A$ is disconnected, say $\text{Spec } A = V(I) \amalg V(J)$ where I and J are proper ideals. Since an element e is idempotent if and only if $e(1 - e) = 0$, what we are looking for is a decomposition of 1 as $e + f$, where $ef = 0$ and $e, f \neq 1$. Now we know (cf. 3.7(v)) that $V(I + J) = V(I) \cap V(J) = \emptyset$, so $I + J = A$ by 3.7(iii). Thus $e_1 + f_1 = 1$ for some $e_1 \in I, f_1 \in J$. We also know (3.7(iv)) that $V(IJ) = V(I) \cup V(J) = \text{Spec } A$, so every element of IJ is nilpotent by 3.7(ii). In particular, $e_1 f_1$ is nilpotent. I claim now that we can get our desired decomposition $e + f = 1$ by raising both sides of the equation $e_1 + f_1 = 1$ to a high enough power. Indeed, for any integer $n > 0$, we can group the terms in the binomial expansion of $(e_1 + f_1)^{2n-1}$ to obtain $1 = (e_1 + f_1)^{2n-1} = e + f$, where $e \in Ae_1^n \subseteq I$ and $f \in Af_1^n \subseteq J$ (hence $e, f \neq 1$). Since $e_1 f_1$ is nilpotent, we will then have $ef = 0$ for large enough n . $\square$

EXERCISES

- *1. Generalize 3.4 as follows: For any commutative ring A , the idempotents of A are in 1-1 correspondence with the subsets of $\text{Spec } A$ which are both open and closed.
- *2. Generalize 3.5 as follows: Let A be an arbitrary commutative ring and let P be a finitely generated projective A -module. For any integer $i \geq 0$ let $U_i \subseteq \text{Spec } A$ be $\mathcal{R}_{A(P)}^{-1}(i)$. Let $e_i \in A$ be the idempotent corresponding to U_i (cf. exercise 1). Then $e_i = 0$ for all but finitely many i and $R_A(P) = \sum_{i \geq 0} ie_i$.
- *3. In view of exercise 4 of §2, the result of exercise 2 can be stated in the following concrete terms: Let E be an idempotent matrix over a commutative ring A ; then there are idempotent elements $e_i \in A$ such that $\text{tr}(E) = \sum_{i \geq 0} ie_i$. It is natural to ask whether the e_i can be computed explicitly in terms of the entries of the matrix E . The purpose of this exercise is to give an affirmative answer, due to Goldman [1961] (see also Almkvist [1973]), which gives a formula for e_i in terms of the coefficients of the characteristic polynomial of E .
 - (a) Let E be an idempotent $n \times n$ matrix and let $\varphi(t) = \det(1 + tE) = 1 + \text{tr}(E)t + \dots + \det(E)t^n$. Let P be the corresponding projective module $\text{Im}(E)$, and let $(e_i)_{i \geq 0}$ be as in exercise 2. Show that $\varphi(t - 1) = \sum_{i \geq 0} e_i t^i$. [Hint: At a prime ideal where P has rank i , both sides of this equation are equal to t^i .]

(b) Writing $\varphi(t) = \sum_i a_i t^i$, deduce that

$$e_i = \sum_j (-1)^j \binom{i+j}{i} a_{i+j}.$$

If $n = 2$, for example, then $e_2 = \det(E)$, $e_1 = \operatorname{tr}(E) - 2 \det(E)$, and $e_0 = 1 - \operatorname{tr}(E) + \det(E)$.

4 Ranks over Group Rings; Swan's Theorem

For simplicity we will confine ourselves to *integral* group rings $\mathbb{Z}\Gamma$, although it will be obvious that some of what we do in this section extends to more general group rings $k\Gamma$.

In order to understand the Hattori–Stallings rank $R(P)$ (also denoted $R_\Gamma(P)$) over $\mathbb{Z}\Gamma$, we must first understand the group $T(\mathbb{Z}\Gamma)$. This group is the quotient of the additive group of $\mathbb{Z}\Gamma$ by the subgroup generated by the commutators $[\gamma, \gamma'] = \gamma\gamma' - \gamma'\gamma$ ($\gamma, \gamma' \in \Gamma$). Now the commutators of this form are precisely the elements $\gamma\gamma_1\gamma^{-1} - \gamma_1$ ($\gamma, \gamma_1 \in \Gamma$); for we have $[\gamma, \gamma'] = \gamma(\gamma'\gamma)\gamma^{-1} - (\gamma'\gamma)$ and $\gamma\gamma_1\gamma^{-1} - \gamma_1 = [\gamma, \gamma_1\gamma^{-1}]$. It follows easily that $T(\mathbb{Z}\Gamma)$ can be identified with the free abelian group on the conjugacy classes of elements of Γ . Thus any element of $T(\mathbb{Z}\Gamma)$ has the form

$$\sum_{\gamma \in \mathcal{C}} t(\gamma) \cdot [\gamma],$$

where $\mathcal{C}$ is a set of representatives for the conjugacy classes in Γ , $[\gamma]$ denotes the conjugacy class of γ , and $t: \Gamma \rightarrow \mathbb{Z}$ is a function which is constant on each conjugacy class and is zero for almost all conjugacy classes. In concrete terms, then, the trace of a matrix over $\mathbb{Z}\Gamma$ is obtained by summing the diagonal elements and grouping together the terms corresponding to conjugate group elements.

Now let P be a finitely generated projective $\mathbb{Z}\Gamma$ -module. Its Hattori–Stallings rank $R_\Gamma(P) \in T(\mathbb{Z}\Gamma)$ has an expansion

$$R_\Gamma(P) = \sum_{\gamma \in \mathcal{C}} R_\Gamma(P)(\gamma) \cdot [\gamma].$$

Thus $R_\Gamma(P)$ can be viewed as a family of integers $R_\Gamma(P)(\gamma)$ ($\gamma \in \mathcal{C}$) associated to P . From this point of view, the restriction formula 2.4 takes the following form:

(4.1) Proposition. *Let Γ be a group and Γ' a subgroup of finite index. For any $\gamma \in \Gamma'$ there is an integer $n(\gamma) > 0$ such that for every finitely generated projective $\mathbb{Z}\Gamma$ -module P ,*

$$R_\Gamma(P)(\gamma) = n(\gamma) \cdot R_{\Gamma'}(P)(\gamma).$$

Moreover, $n(1) = (\Gamma: \Gamma')$.